Republic of Liberia

Health Equity and Social Determinants of Health Assessment

October 30, 2023

Foreword

It is with great privilege and a profound sense of responsibility that I introduce this Health Equality Assessment Report. In an era defined by remarkable advances in medical science and technology, we must constantly evaluate our progress not only in development of novel treatments but also in ensuring equitable access to healthcare for all.

Health is a fundamental human right, a cornerstone of well-being, and a reflection of societal values. Yet, disparities persist in healthcare access and outcomes, often disproportionately affecting vulnerable and marginalized communities. As we embark on this journey of assessment, we confront the stark reality that health inequalities persist, and in some cases, have deepened.

This report represents an earnest endeavor to uncover the root causes of health disparities, to measure the extent of inequality, and to propose evidence-based solutions to bridge these divides. It has been crafted through rigorous research, collaboration, and a commitment to fostering equitable healthcare system.

We must acknowledge that the pursuit of health equality is not a one-size-fits-all endeavor. Our world is characterized by a rich tapestry of cultures, socio-economic conditions, and healthcare infrastructures. Thus, this report provides Liberia's perspective, recognizing that the fight for health equity is a shared responsibility transcending borders.

In these pages, you will find an analysis of the systemic challenges that hinder progress towards health equality, but you will also encounter stories of resilience, innovation, and compassionate care that inspire hope. This report is not only a diagnosis of our current state but a call to action. It is a roadmap towards a future where every Liberian, regardless of their background or circumstance, can access quality healthcare and live a healthy life.

I extend my heartfelt gratitude to the dedicated data collectors and producers, healthcare professionals, and advocates who have contributed to this report. Your unwavering commitment to the cause of health equality shines through in these pages. I also urge policymakers, healthcare leaders, and development partners heed the findings and recommendations presented here.

Let this report serve as a catalyst for change, a blueprint for equitable healthcare systems, and a reminder that our shared humanity calls us to stand together in the pursuit of health for all.

Together, we can build a healthier, fairer Liberia.

Hon. Wilhemina Jallah, MD
Minister of Health
Republic of Liberia

Acknowledgements

We would like to express our sincere gratitude to all those who contributed to the Health Equality Assessment Report. This endeavor would not have been possible without the dedication, expertise, and support of the following individuals and organizations.

We extend our heartfelt appreciation to Assistant Minister Chea Sanford Wesseh, the consultant, for his commitment and tireless efforts in data collection, analysis, and reporting.

We are deeply thankful to the World Health Organization (WHO) for their guidance and valuable contribution during the conduct of the Health Equality and Equity Assessment. We are grateful for the financial resources provided to conduct the assessment, and the report validation workshop.

We extend our thanks to the healthcare workers, especially, County Health Officers (CHOs), and their Human Resource Officers (HR) that generously shared data and insights to inform our assessment.

We are grateful to everyone who participated in the data collection, reviewed and validated the draft report and shared their experiences and perspectives during the report validation workshop. Our heartfelt thanks to those who provided valuable feedback and contributed to the refinement of this report.

Lastly, the Government and her development partners are committed to addressing the precarious conditions face by counties, districts, and communities affected by health inequalities and deprivations. Your voices have been heard, and your experiences have been a driving force in our pursuit for health equality and equity.

WHO remains committed to addressing health disparities and inequalities and is working towards a healthier, and equitable future for all.

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List of Acronyms and abbreviation

AIDS	Acquired Immunodeficiency Syndrome
ANC	Antenatal Care
ARV	Antiretroviral
ART	Antiretroviral therapy
BEmOC	Basic Emergency Obstetric Care
CEmOC	Comprehensive Emergency Obstetric Care
CSOs	Civil Society Organizations
DHS	Demographic and Health Survey
DHIS2	District Health Information System version 2
EVD	Ebola Virus Disease
FY	Fiscal Year
HIV	Human Immunodeficiency Virus
HMIS	Health Management Information System
HHFA	Harmonized Health Facility Assessment
HIES	Households Income and Expenditure Survey
HR	Human Resources
HRH	Human Resources for Health
IPF	In-patient Facility
LDHS	Liberia Demographic and Health Survey
NGOs	Non-Governmental Organization
MOH	Ministry of Health
OTP	Out-patient Therapeutic Program
UHC	Universal Health Coverage
UNICEF	United Nations Children’s Education Fund
WHO	World Health Organization

Definition of Key Concepts

Health Inequality: observable health status differences between subgroups of a population; and can be measured and monitored.

Health Inequity: avoidable differences (rates) in health between subgroups of a population or geographical locations due to differences in social, economic, political, residential environment or healthcare resources.

Social Determinants of Health: circumstances in which people are born, grow up, live, work and age, as well as the systems put in place to deal with illness. These circumstances in turn are shaped by a wider set of forces which include the distribution of economics, power, politics and resources at national and local levels.

Equity-based Interventions: seek to improve health outcomes in subgroups of a population or geographical locations that are disadvantaged, vulnerable and deprived while improving the overall situation.

Health systems: as defined by WHO describes "all the activities whose primary purpose is to promote, restore, or maintain health."

Executive Summary

Liberia's health system is fragile due to a limited period of recovery after the Ebola Virus Disease Outbreak in 2014-2015, followed by the Corona Virus Disease pandemic in 2020-2023. These health emergencies coupled with weak governance and leadership, inadequate investment, a demotivated health workforce and frequent stock out of essential drugs and medical supplies further weaken the health system and deprive inhabitants of access to quality healthcare.

The World Health Organization commissioned a health equity analysis to determine the level of inequality and deprivation within the health sector. The intent of the equality assessment is to document geographic locations with health deprivation and/or limited access to basic healthcare. Also, the assessment seeks to unearth counties with equality issues and how to mitigate these challenges. The health sector is faced with systemic challenges which include the following:

- Limited access to healthcare services, especially in rural areas.
- Acute shortage of healthcare workers, including specialist doctors, certified midwives, and laboratory technicians.
- Inadequate healthcare infrastructure and medical equipment.
- High maternal and childhood mortality rates,
- Prevalence of diseases like malaria, HIV/AIDS and tuberculosis, and
- Limited health security capacity

The specific objectives of the assessment are as follows:

(a) To conduct health equity analysis and identify social determinants of health.
(b) To develop a comprehensive health equity analysis report.
(c) To identify opportunities to address avoidable health inequities.
(d) To improve the delivery of health services based on the identified inequalities and
(e) To utilize key findings and recommendations to inform evidence-based policy and strategic direction on addressing social determinants of health equity in Liberia.

A country-driven survey and desk review were conducted to assess and analyze different aspects of health equity that would address the specific objectives. The methodology involves a structured approach to evaluating and reporting on health inequalities or disparities across counties.

A significant portion of Liberia's population lives below the poverty line, leading to limited access to healthcare, nutritious food, and safe housing. Little over half of Liberians (50.9%) are living in abject poverty; 39.1% are experiencing food poverty, and 16.5% extreme poverty (LISGIS, HIES).

The impact of high unemployment rates on financial stability and access to necessary health services is deeply concerning. Sadly, Liberia is experiencing this issue on a large scale, with a significant portion of the population aged 15 and above currently unemployed. Disturbingly, according to the 2019/20 Liberia Demographic and Health Survey, 39% of women and 19% of men are unemployed with only 4% of Liberians having health insurance.

Health facilities are unevenly distributed with some health districts having a single health facility serving over 5,000 people.

In Liberia, there is an insufficiency of hospital beds. Due to inadequate beds, patients are either prematurely discharged or refused admission. Hospital beds are unfairly allocated and distributed to counties and health facilities. The patient-to-hospital beds ratio shows that for every 10,000 people in Maryland County there are two beds, while in Gbarpolu and Grand Kru Counties, there are four beds, and nationally, there are 17 hospital beds per every 10,000 people.

Nationally, 70.3% of inhabitants of Liberia have access to healthcare. Counties with less than 50% of their population having access to health care are, Rivercess (40.4%), Gbarpolu (49.4%), and Grand Bassa. However, Montserrado has the highest access at 87.6%, followed by Margibi (71%) Counties.

Health services utilization is very low in Liberia. Over the past 5years (2018-2022), utilization has been below one visit per capita which is an 83% reduction of the WHO recommended five visits per capita per year. The highest service utilization was found in Grand Kru (1.06 per capita), River Gee (1.03 per capita), and Montserrado (0.97 per capita) counties. The lowest service utilization was reported from Gbarpolu (0.52 per capita), Grand Cape Mt (0.69 per capita), and Bong (0.72 per capita) Counties (MOH HMIS).

Liberia is among countries with a very high infant mortality rate. The current rate is 63 deaths per 1,000 live births. The highest infant mortality rates are found in Grand Cape Mount County (116 deaths per 1,000 live births), and Sinoe County (112 deaths per 1,000 live births). Also, the lowest infant deaths were recorded in Bong County (53 deaths per 1,000 live births), and Bomi County (53 deaths per 1,000 live births) (DHS 2019/20).

The number of adult deaths varies across Liberia with Grand Kru County (20 deaths per 1,000 population), and Lofa County (20 deaths) having the highest number of deaths per 1,000 population, compared to Montserrado (9) and Margibi (10) Counties with the lowest death rates.

Mortality is increasing among all sub-groups of the population. For instance, over the past three years, on average five maternal deaths and 15 neonatal deaths have been reported weekly. To address this appalling and undesirable situation, the government and its development partners need to invest in the quality of maternal and newborn health services, ensure the availability of essential drugs and life-saving commodities, improve referral services, and expand CEmOC facilities and their quality.

There is an acute shortage of Laboratory technicians in Liberia. The health workforce mapping data displays significant variations from one county to another. The Lab technician density ratio shows less than 1 Lab technician per 100,000 population in Grand Bassa, Bomi, and Sinoe Counties. On the other hand, there are 12.7 Lab technicians and 12.1 in Maryland and Rivercess Counties respectively.

Interventions to address health equality and inequality involve policies and initiatives that aim to reduce disparities, improve healthcare access, and promote social determinants of health. These strategies must include providing affordable healthcare, improving income, and education, and addressing systemic factors that contribute to health disparities.

1.0 Introduction and Background

1.1 Country's Profile

The Republic of Liberia, which is situated on the west coast of Africa, is a nation with a rich and complex history. It is known for being one of the few African countries to that were never formally colonized by European powers. The country was founded by African Americans and freed slaves from the United States in the 19th century. Liberia's history, culture, and society are deeply influenced by this American connection.

Liberia's capital is Monrovia and has a land area of 111,369 square kilometer and an estimated population of 5.2 million[1]. The official language is English and the Liberian Dollar is the official currency.

Geography

Liberia's diverse landscape includes lush tropical rainforests, extensive coastline along the Atlantic Ocean, rolling plains, and mountainous regions. The country's interior features dense forests and numerous rivers, including the iconic Cavalla River. Liberia is subdivided into 15 counties, 73 electoral districts and 136 administrative districts.

History

Liberia's history is marked by resettlement of freed African-American and Afro-Caribbean slaves in the 1820s. The capital city, Monrovia, was named after the U.S. President James Monroe, who supported the establishment of the colony. Liberia declared its independence on July 26, 1847, becoming the first African nation to proclaim sovereignty. Liberia experienced a devastating 14 years of civil war that shattered the health system, growth and development. Also, in 2014, Liberia faced one of the deadliest Ebola Virus Disease (EVD) epidemic that resulted to over 4,000 deaths and 10,000 infections.

Politics

Liberia is a republic with a presidential system of government. The President serves as both the head of state and head of government. The country has a multi-party system, with periodic elections. The President can only serve for two terms which is equivalent to twelve years.

Economy

Agriculture, mining, and forestry are significant contributors to Liberia's economy. The country possesses valuable natural resources, including iron ore, diamonds, rubber, and timber. However, the economy has faced challenges, including periods of civil conflict and instability. The EVD epidemic and the COVID-19 pandemic impacted revenue collection and the economy.

[1] 2022 National Population and Housing Census

Demography

Liberia's population is almost evenly split between males and females, with 50.4% being male and 49.6% being female[2]. Eighty-five percent (85%) of the population is Christian and 63.5% of the population between the ages of 15-49 are literate. Unfortunately, there is still a significant gender disparity in education[3], with 41% of females and 30% of males aged 6 and older having no formal education. Access to improved sources of drinking water and sanitation are also skewed towards urban areas, with 84% of households having access to improved drinking water and only 47% having access to improved sanitation[4].

On a positive note, Liberia has made progress in reducing maternal mortality, with a 31% decline over the course of 6-7 years[5]. Additionally, life expectancy, child health, and maternal indicators have all improved over the same period. The contraceptive prevalence rate has increased, skilled birth attendance has increased, and childhood immunization and nutrition have improved. While there is still progress to be made, it is encouraging to see positive changes happening in Liberia.

Contraceptive prevalence rate increased from 19% in 2013 to 24% in 2019/20, skilled birth attendance swelled from 61% in 2013 to 84% in 2019/20, institutional delivery increased from 56% in 2013 to 80% in 2019/20. The proportion of children who received all basic vaccination increased from 55% in 2013 to 65% in 2019/20. Additionally, stunting declined from 33% in 2013 to 30% in 2019/20 while childhood mortality plummeted from 42 deaths per 1,000 live births in 2013 to 33 deaths in 2019/20[6].

1.2 Liberia Health System

According to the World Health Organization, health is "a state of complete physical, mental and social well-being and not merely the absence of disease and infirmity." Liberia’s health system has faced significant challenges over the years but has also shown resilience and improvement efforts.

Healthcare Infrastructure

Liberia’s healthcare infrastructure suffered greatly during the civil war that ended in 2003. Many healthcare facilities were destroyed, and the country’s healthcare system was left in disarray. Efforts were made to rebuild and expand the healthcare infrastructures, with the support of international organizations. There are 935 health facilities across Liberia, inclusive of both public (50.5%) and private (49.5%)[7]. Annex C provides the number of health facilities by county, type and ownership.

Healthcare Delivery

Healthcare services in Liberia are provided through a mix of public and private healthcare facilities. Public healthcare is managed by the Ministry of Health and supported by

[2] Liberia Population and Housing Census, 2022
[3] Liberia Demographic and Health Survey, 2019/20
[4] Ibid
[5] Ibid
[6] Ibid
[7] Ministry of Health, Health Management Information System database, 2022

international donors and health partners. Private healthcare providers, including non-governmental organizations and faith-based organizations, also play a significant role in delivering healthcare services.

Health Financing

The government of Liberia, with the support of international partners, has been working on improving healthcare financing mechanisms such as performance-based healthcare financing (PBF), Fixed Assets Reimbursement Arrangement (FARA), Revolving Drugs Fund (RDF) and healthcare insurance scheme. These efforts aim to increase public funding for healthcare and reduce the financial burden on individuals, especially the poor.

Human Resources for Health (HRH)

There are approximately 24,423 health workers inclusive of 10,344 community health workers (535-CHSS & 4,969-CHAs), that provide healthcare to 5.2 million people (Liberia Population and Housing Census, 2022). Of this number, 9,758 (50.5%) are civil servants (MOH payroll, 2023) and 46.2% are community health workers (CHAs, TTM, CHSS, etc). This health workforce includes 323 medical doctors, 467 physician assistants, 3,701 nurses, 1,286 midwives and 347 Laboratory technicians among others.

Health Policies

Liberia has developed various health policies and initiatives to address its healthcare challenges. The National Health Policy and Plan, along with the Essential Package of Health Services (EPHS), guide the country's efforts to improve healthcare delivery.

Partnerships and International Support

Liberia collaborates with international organizations and donors to strengthen its health system. These partnerships provide financial support, technical assistance, and capacity-building programs. Development partners contribute over 50% of healthcare financing for service delivery and technical assistance for the development of health policies and strategies.

Healthcare Challenges

Liberia has faced challenges such as civil wars (1989-2003), Ebola outbreak (2014-2016) and the Corona Virus Disease (COVID-19) pandemic that shattered the delivery of quality healthcare and restricted the growth and development of the sector. The systemic challenges of the health sector are:

- Limited access to healthcare services, especially in rural areas.
- Acute shortage of healthcare workers, including specialist doctors, certified midwives and laboratory technicians.
- Inadequate healthcare infrastructure and medical equipment.
- High maternal and childhood mortality rates,
- Prevalence of diseases like malaria, HIV/AIDS and tuberculosis, and
- Limited health security capacity

1.3 Objectives

The overall objective of the assessment in Liberia is to document and highlight health inequalities and inequities. The specific objectives are as follows:

(f) To conduct health equity analysis and identify social determinants of health.
(g) To develop a comprehensive health equity analysis report.
(h) To identify opportunities to address avoidable health inequities.
(i) To improve the delivery of health services based on the identified inequalities and
(j) To utilize key findings and recommendations to inform evidence-based policy and strategic direction on addressing social determinants of health equity in Liberia.

1.4 Methodology

A country-driven survey and desk review was conducted to assess and analyze different aspects of health equity that would address the specific objectives. The methodology involved a structured approach to evaluating and reporting on health inequalities or disparities across counties.

1.4.1 Desk review

A comprehensive review of existing research and literature related to health inequalities in the target population was conducted. The existing research and Literature reviewed included the Liberia Demographic and Health Survey (LDHS), the 2022 National Population and Housing Census provisional results, the health sector resource mapping report, the Harmonized Health Facility Assessment (HHFA), the health management information platform (HMIS/DHIS2), among others.

1.4.2 Data Collection

A simple excel template was developed to capture the health workforce data by health facilities across Liberia. After the development of the template, an orientation meeting was held for Ministry of Health Human Resource (HR) Officers (data collectors) assigned in each county to collect HR data from both public and private health facilities including community healthcare workers' data. The only primary data that was collected during the assessment was the human resources for health data. Additionally, the health information system data base (DHIS2) that captures routine health information was used as the primary source for service utilization data.

1.4.3 Data Analysis

The assessment measured socioeconomic-related inequalities in health variables by using rates, ratios and percentages. Absolute difference, relative difference (ratio), were used to analyse and describe health equalities.

2.0 Social Determinants of Health

2.1 Introduction

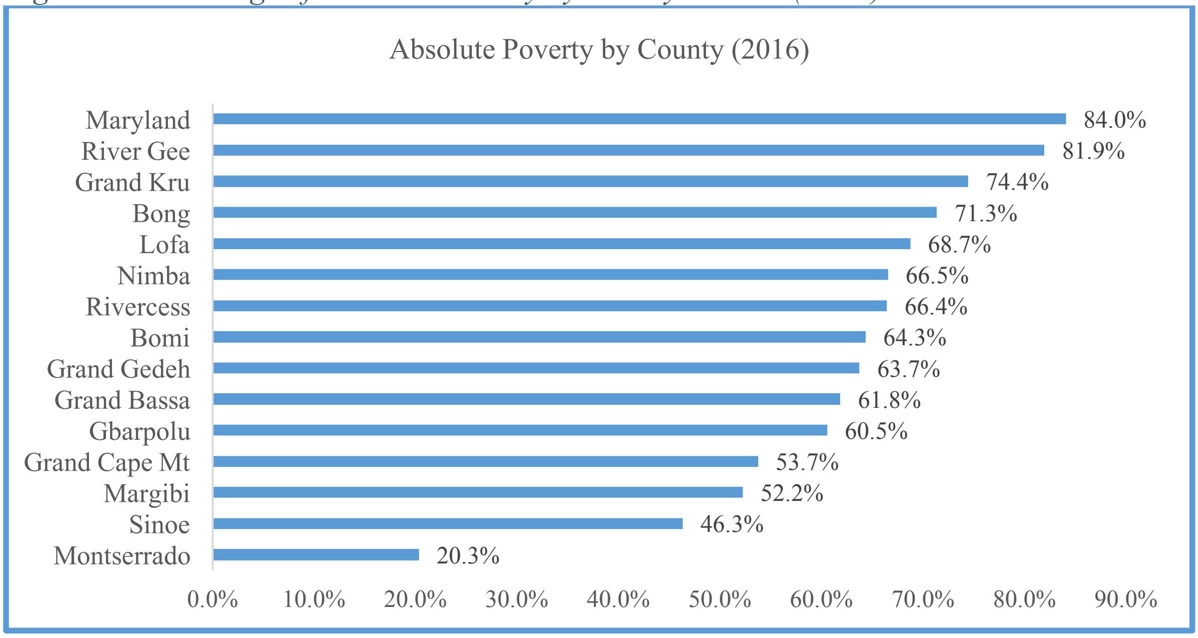
Liberia, a West African nation, faces a complex interplay of social determinants that significantly impact the health and well-being of its population. These social determinants encompass various economic, political, cultural and environmental factors:

2.2 Economic Factors

2.2.1 Poverty

Significant portion of Liberia's population lives below the poverty line, leading to limited access to healthcare, nutritious food and safe housing. Little over half of Liberians (50.9%) are living in abject poverty; 39.1% are experiencing food poverty, and 16.5% extreme poverty[8]. Figure 1 below shows that poverty is extremely high in three of the Southeastern counties of Maryland (84%), River Gee (81.9%), and Grand Kru (74.4%), while Montserrado has the lowest poverty rate of 20.3%.

Figure 1: Percentage of Absolute Poverty by County in 2016 (HIES)



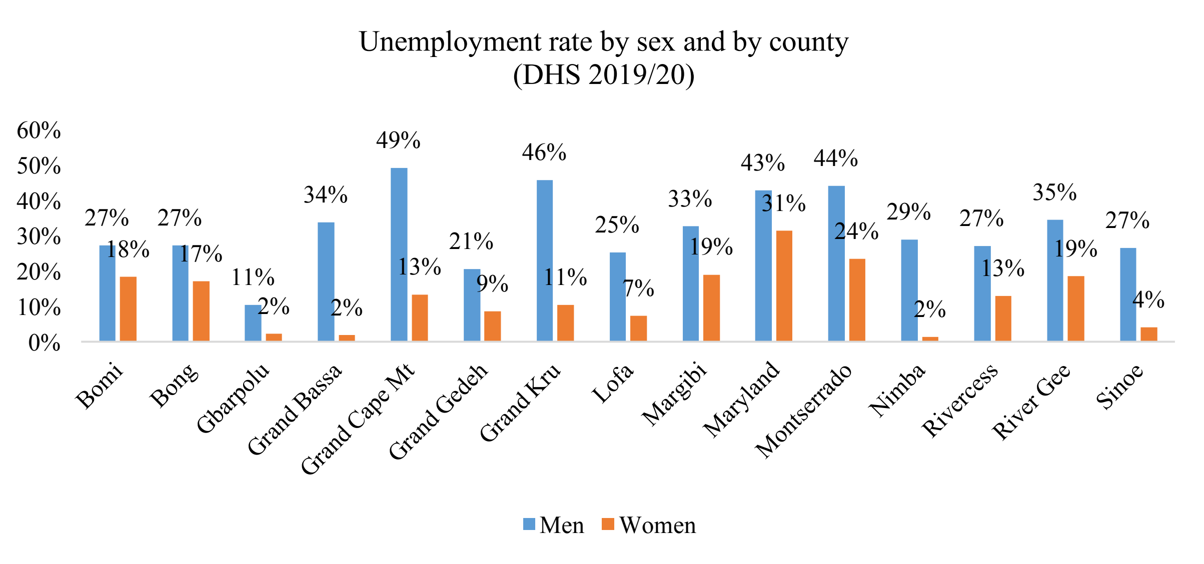
2.2.2 Unemployment

The impact of high unemployment rates on financial stability and access to necessary health services is deeply concerning. Sadly, Liberia is experiencing this issue on a large scale, with a significant portion of the population aged 15 and above currently unemployed. Disturbingly, according to the 2019/20 Liberia Demographic and Health Survey, 39% of women and 19% of men are unemployed. This issue is affecting women more than men, with one-third of the counties in Liberia reporting high unemployment rates. These counties include Grand Cape Mount (49% of women), Grand Kru (46% of women), Montserrado (44% of women), Maryland (43% of women), and River Gee (35% of women). It is crucial that the government

[8] Household Income and Expenditure Survey, 2016

and her development partners find solutions to this problem to ensure that all Liberians can have the opportunity to thrive. Figure 2 presents unemployment rates by sex and by county.

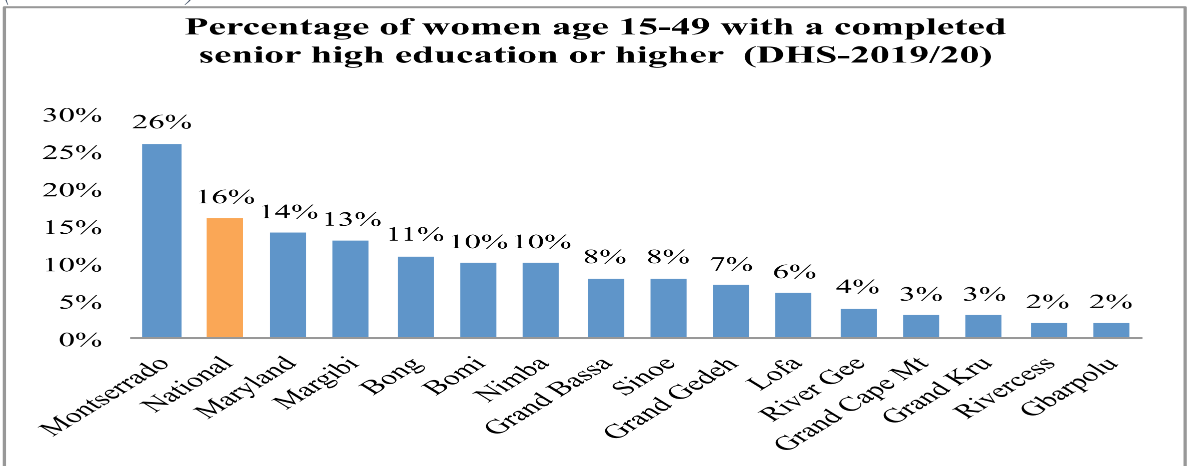
Figure 2: Unemployment Rate by Sex and by County (DHS 2019/20)



2.3 Education and Literacy

It is important to note that limited access to quality education can have a significant impact on an individual's health literacy and awareness, which can lead to poor health decision-making. Unfortunately, many Liberians do not have formal education, with a significant portion of females and males aged six years and older having no formal education or only some elementary education. According to the recent DHS, some counties have lower rates of women aged 15-49 completing senior high education or higher, which could further exacerbate the issue. It is crucial that we work together to find ways to improve access to education for all Liberians to ensure that they have the tools they need to make informed decisions about their health. Figure 3 depicts the percentage of women with a completed senior high education or higher.

Figure 3: Percentage of women age 15-49 with a completed senior high education or higher (DHS 2019/20).

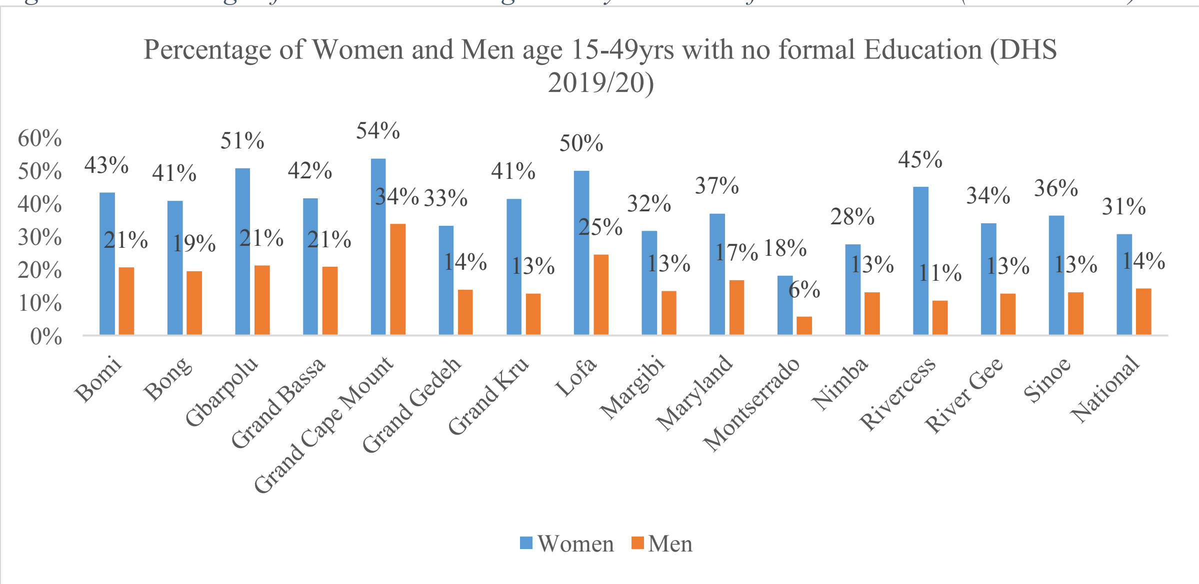


2.3.1 Formal Education

Formal education refers to structured, organized, and systematic learning that takes place in institutions such as schools, colleges, and universities. It naturally follows a prescribed curriculum and is guided by teachers or instructors. Formal education is essential for acquiring knowledge and skills in various subjects and is often a prerequisite for many careers and professional opportunities. It normally includes primary, secondary, and higher education levels.

In Liberia, 31% of women and 14% of men age 15-49 have no formal education. The proportion of men and women with no formal education varies from county to county. However, Grand Cape Mt County (54%), Gbarpolu (51%) and Lofa (50%) have the highest proportion of women with no formal education. Figure 4 depicts the percentage of women and men age 15-49 years with no formal education.

Figure 4: Percentage of women and men age 15-49yrs. with no formal education (DHS 2019/20)

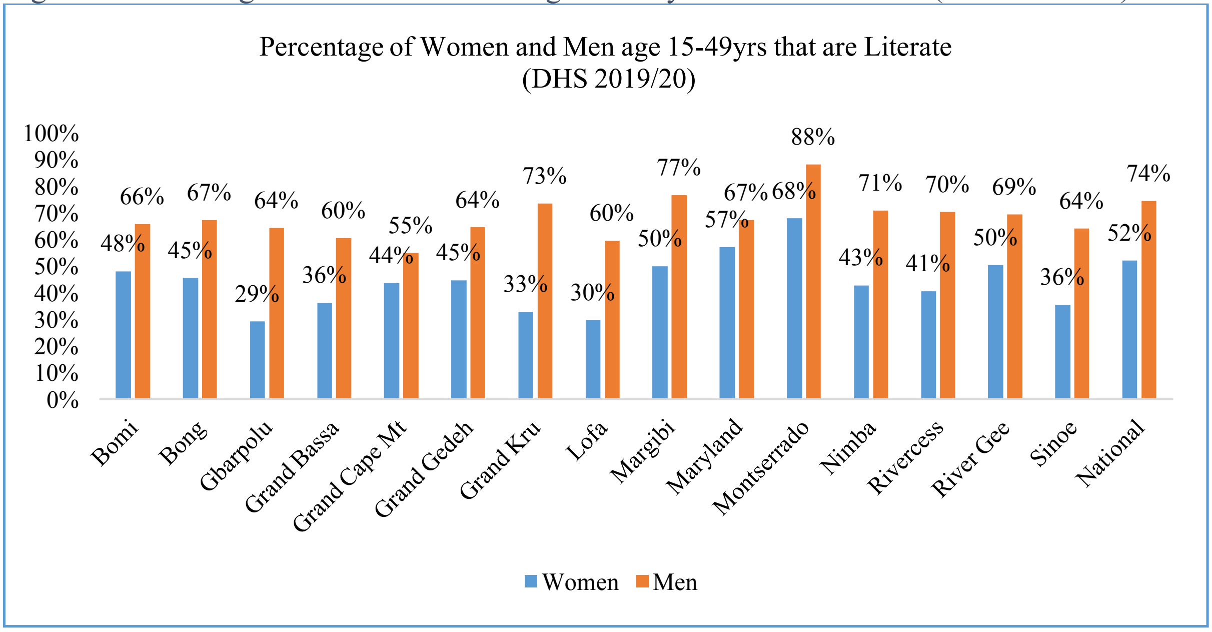


2.3.2 Literacy

Literacy is the ability to read and write, and it is a fundamental skill that enables individuals to access information, communicate, and participate in society. It plays a crucial role in education, personal development, and economic opportunities.

In Liberia, men are most likely to be literate then women. Little over half (52%) of women age 15-49, and 74% of men the same age, are literate. Literacy varies from one county to another and by sex. Gbarpolu (29%), Lofa (30%) and Grand Kru (33%) have the lowest female literacy rate. Figure 5 displays the percentage of women and men age 15-49 who are literate.

Figure 5: Percentage of women and men age 15-49 yrs. that are Literate (DHS 2019/20)



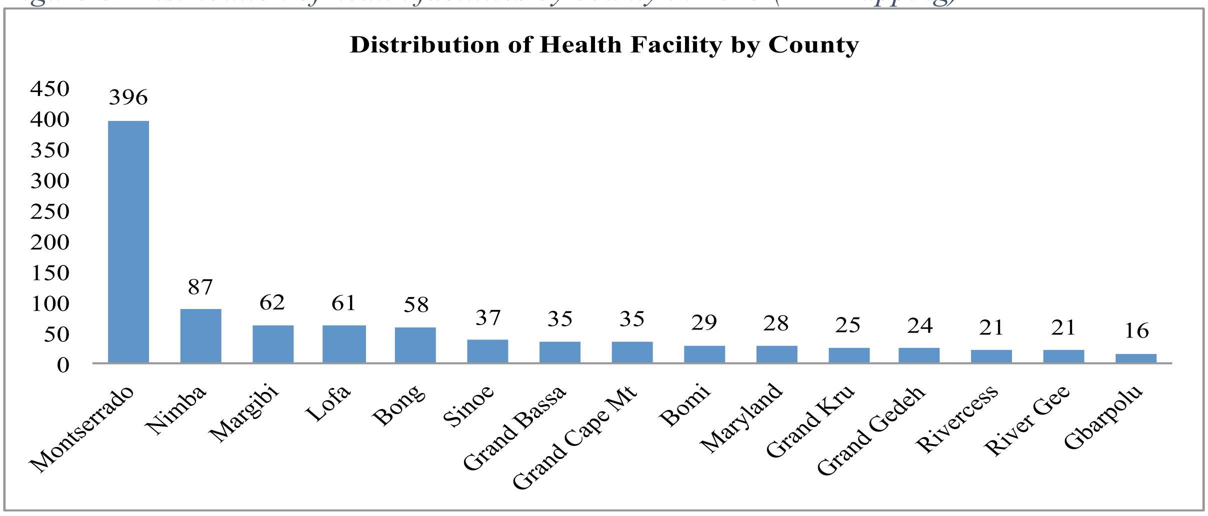
2.4 Healthcare System

Liberia's healthcare infrastructure faces challenges, with trained healthcare professionals, and essential medication. Geographic barriers and transportation issues hinder access to healthcare particularly in rural areas.

2.4.1 Health Facility

There are 1,064 health facilities across Liberia that provide health services to 5.2 million people. Health facilities are unevenly distributed with some health districts having a single health facility serving over 5,000 people. Figure 6 depicts the number of health facilities by county.

Figure 6: Distribution of health facilities by county in 2023 (HF Mapping)



2.4.2 Human Resources for Health (HRH)

Liberia has over 24,583[9] healthcare workers that provide health services to the population. Community health workers account for the highest (46%) cadre of health workers, followed by nurses (15.4%) and midwives (5.3%). Health workers' distribution is skewed towards counties that are either urbanized or with social amenities. Table 1 below presents the distribution of health workers by county.

Table 1: Distribution of health workers by county in 2023 (MOH HR Mapping)

#	County	Health Facility	Medical Doctors	Physician Assistant	Nurses	Certified Midwife	Pharmacist	Lab Tech	Total	Lab Aid	Nurse Aid	Dispenser	CHW	Others	Total HW
1	Bomi	29	7	12	160	41	3	2	225	38	76	33	321	188	881
2	Bong	58	29	17	384	132	9	25	596	27	171	76	2,387	428	3,685
3	Gbarpolu	16	4	1	67	24	2	2	100	10	34	22	462	88	716
4	Grand Bassa	35	16	11	185	30	5	8	255	22	55	26	823	183	1,364
5	Grand Cape Mt	35	5	18	118	28	4	6	179	14	59	38	866	175	1,331
6	Grand Gedeh	24	5	5	64	81	1	11	167	29	62	39	433	189	919
7	Grand Kru	25	4	17	50	55	3	7	136	20	28	29	488	224	925
8	Lofa	61	11	13	227	119	4	6	380	35	157	80	796	334	1,782
9	Margibi	62	55	53	413	103	10	34	668	72	226	79	405	507	1,957
10	Maryland	28	11	16	135	62	2	9	235	18	55	36	559	267	1,170
11	Montserrado	396	132	274	1262	365	28	162	2,223	209	631	210	793	1,407	5,473
12	Nimba	87	43	22	497	114	12	62	750	102	305	85	2,186	789	4,217
13	Rivercess	21	6	8	97	40	5	11	167	10	24	23	296	242	762
14	River Gee	21	5	14	66	47	3	10	145	13	41	25	355	214	793
15	Sinoe	37	2	5	72	60	2	1	142	42	66	48	125	175	598
	Total	*935*	*335*	*486*	*3,797*	*1,301*	*93*	*356*	6,368	*661*	*1,990*	*849*	*11,295*	*5,410*	*24,583*

2.4.3 Healthcare Financing

Since 2020/21 fiscal year (FY), the national budget continues to increase. The budget increased from US$ 570.1 million in FY 2020/21 to US$ 782.9 million in 2023. On the contrary, external funding to the health sector continue to decline from US$ 339 million in FY 2017/18 to US$ 147.60 million in 2023[10]. Also, the national budget allocated to the health sector continues to fluctuate since 2019/20 FY. The health financing resource mapping exercise revealed that only 23% of the health sector allocated resources in 2023 was on budget, while 77% was of-budget[11]. Table 2 below shows a summary of the health sector financial resources since FY 2107/18.

Table 2: Summary of Health Sector Financial Resources Trend

Fiscal Year	*2017/18*	*2018/19*	*2019/20*	*2020/21*	*2022*	*2023*
Total GOL National Budget (US$ in Million)	$563.50	$570.10	$532.90	$570.10	$786.50	$782.90
Total External Health Sector Resource (US$ in Million)	$339.00	$228.30	$183.20	$111.10	N/A	$147.60
Total GOL Health Budget (US$ in Million)	$76.90	$82.00	$69.20	$70.40	$78.33	$74.20
% On-budget resources	51%	71%	77%	62%	N/A	23%
% Off-budget resources	46%	29%	23%	38%	N/A	77%
% of Health in GOL budget	14%	14%	14%	12%	10%	9%

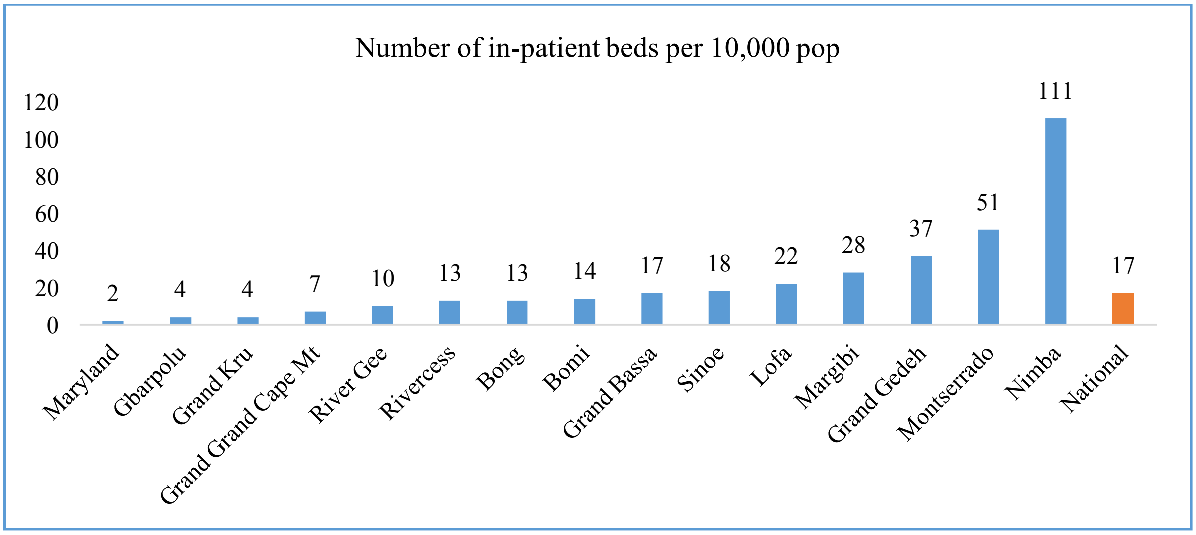
2.4.4 Hospital beds

In-patient beds are essential components of healthcare facilities, such as hospitals and health centers, where patients receive medical care and treatment. These beds are designed to provide comfort, support, and safety for patients during their stay.

[9] MOH Health workers mapping, 2023
[10] Liberia National Budget 2023
[11] MOH Healthcare Financing Resource Mapping exercise, 2023

In Liberia, there is insufficiency of hospital beds. Hospital beds are unfairly allocated and distributed to counties and health facilities. The patient to hospital beds ratio shows that for every 10,000 people in Maryland County there are two beds, while in Gbarpolu and Grand Kru Counties, there are four beds, and nationally, there are 17 hospital beds per every 10,000 people[12]. Figure 7 presents the number of hospital beds per 10,000 population by county.

Figure 7: Number of hospital beds per 10,000 population by County in 2021 (HHFA)

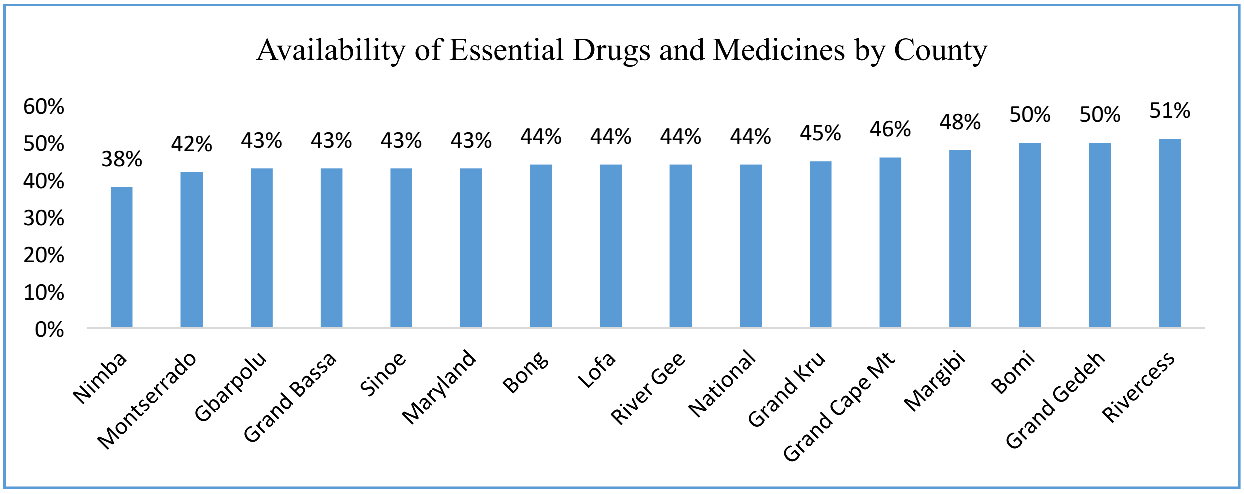


2.4.5 Drugs and Medical Supplies

Overall, 44% of the health facilities in Liberia had essential tracer medicines to provide services with at least 66% of them with ampicillin and ceftriaxone (78%) injections available in stock. Amoxicillin syrup was available in almost three quarters (72%) of the health facilities while, Gentamicin injection was available in stock in 78% of the health facilities. Most important, in the management of diarrhoea, Oral rehydration salts sachets (94%), and Zinc Sulphate tablets or syrups (82%) were the most items available in health facilities, while in management of labour, Oxytocin injection was available in 87% of the health facilities. In the management of acute and chronic conditions such as hypertension, diabetes, Salbutamol inhaler was available in stock in 58% of the health facilities, while insulin regular injection was available in stock in 4% of the health facilities. Figure 8 displays the summary of the availability of essential general medicines.

[12] Harmonized Health Facility Assessment (HHFA), 2021

Figure 8: Availability of Essential Drugs by County, 2021



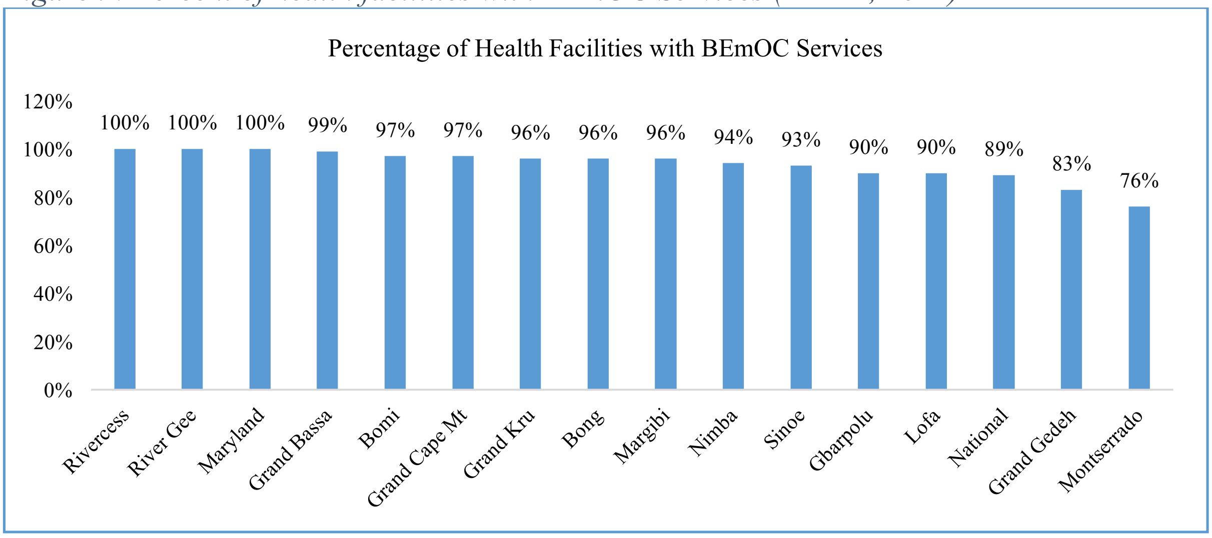
2.4.6 Basic Obstetric and Newborn Services

Basic Emergency Obstetric Care (BEmOC) is a critical component of healthcare services aimed at reducing maternal mortality and ensuring safe childbirth. It encompasses a set of essential interventions and skills that healthcare providers must possess to address emergency situations during pregnancy, childbirth, and the postpartum period.

BEmOC plays a pivotal role in improving maternal and neonatal outcomes, particularly in counties with limited access to healthcare. It serves as a bridge between community-level care and more advanced obstetric services, ultimately contributing to reduction of maternal mortality worldwide.

In Liberia, nearly 90% of health facilities provide BEmOC services. However, Grand Gedeh (83%) and Montserrado (76%) have the lowest proportion of health facilities providing BEmOC Services[13]. Figure 9 shows the percentage of health facilities with BEmOC services.

Figure 9: Percent of health facilities with BEmOC Services (HHFA, 2021)



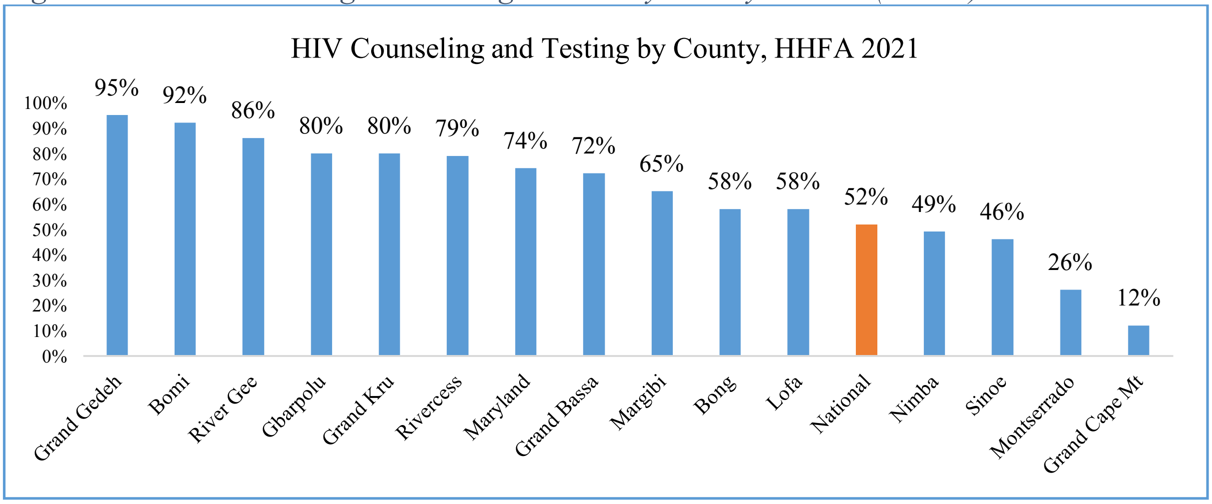
[13] Harmonized Health Facility Assessment, 2021

2.4.7 HIV Counseling and Testing Services

HIV counseling and testing, often referred to as HIV C&T, are crucial components of efforts to prevent, diagnose, and treat HIV/AIDS. In summary, HIV counseling and testing are essential components of HIV/AIDS prevention and care. They do not only diagnose HIV but also provide education, support, and linkage to treatment, ultimately contributing to the reduction of HIV transmission and the well-being of affected individuals.

In Liberia, only half (52%) of the health facilities provide HIV counseling and testing services. The proportion of health facilities that provide HIV counseling and testing services range from 95% in Grand Gedeh to 12% in Grand Cape Mount County[14]. Figure 10 presents the proportion of health facilities that provide HIV counseling and testing services by county.

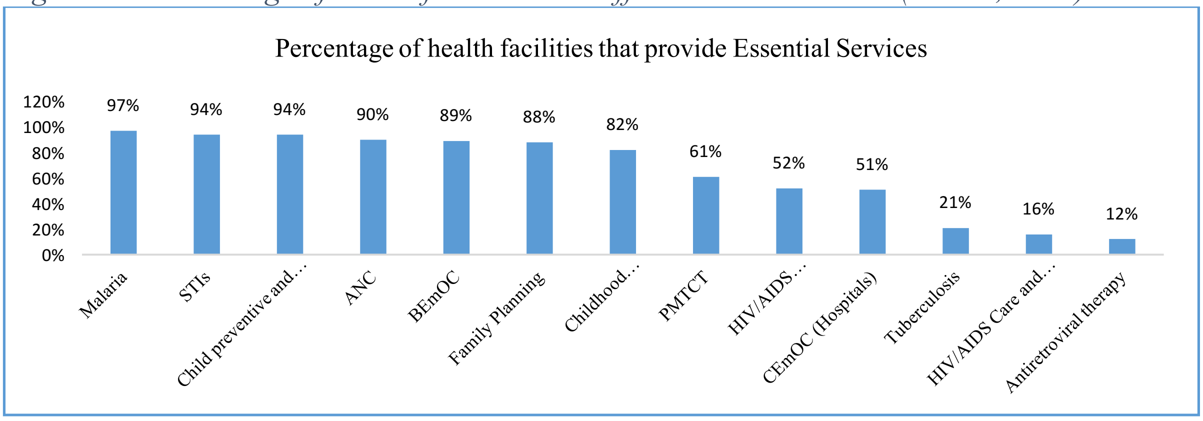
Figure 10: HIV Counseling and Testing Centers by County in 2021 (HHFA)



2.4.8 Essential Health Services

Essential health services are disproportional available across Liberia. For instance, 97% of health facilities offer malaria services, 12% provide TB services and 16% offer HIV/AIDS care and support services[15]. Figure 11 presents the percentage of health facilities that offer selected essential health services in Liberia.

Figure 11: Percentage of health facilities that offer essential services (HHFA, 2021)

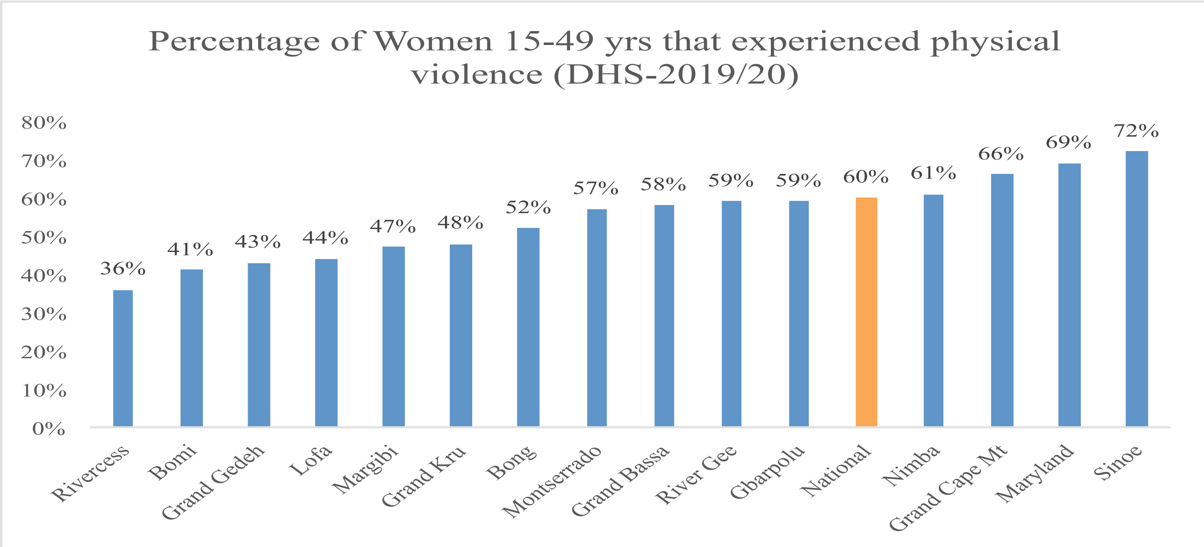


[14] Ibid
[15] HHFA 2021

2.5 Culture and Social Norms

Traditional gender roles and norms can limit women's access to education and healthcare, contributing to maternal and child health disparities. Fifty-five percent of ever married women have experienced emotional, sexual, or physical violence from their current or most recent husband or partner, and 46% have experience such violence in the past 12 months. Counties with widespread domestic violence against women are Sinoe (72%), Maryland (69%) and Grand Cape Mount (66%). Figure 12 displays domestic violence against women by county.

Figure 12: Percentage of ever-married women age 15-49 who have ever experienced physical, sexual or emotional violence committed by their husband or partner (DHIS 2019/20)



Stigma surrounding certain health conditions, such as HIV/AIDS can discourage individuals from seeking care. Pervasive stigma and discrimination in a population can adversely affect both people's willingness to be tested and their adherence to ART. Thus, reduction of stigma and discrimination in a population is an important indicator of success of programs targeting HIV prevention and control. According to the 2019/20 DHS, 53% of women and 50% of men think that children living with HIV should not attend school with children who are HIV negative; 62% of women and 54% of men would not buy fresh vegetables from shopkeeper with HIV. Table 3 below presents the percentage of women and men recently tested for HIV.

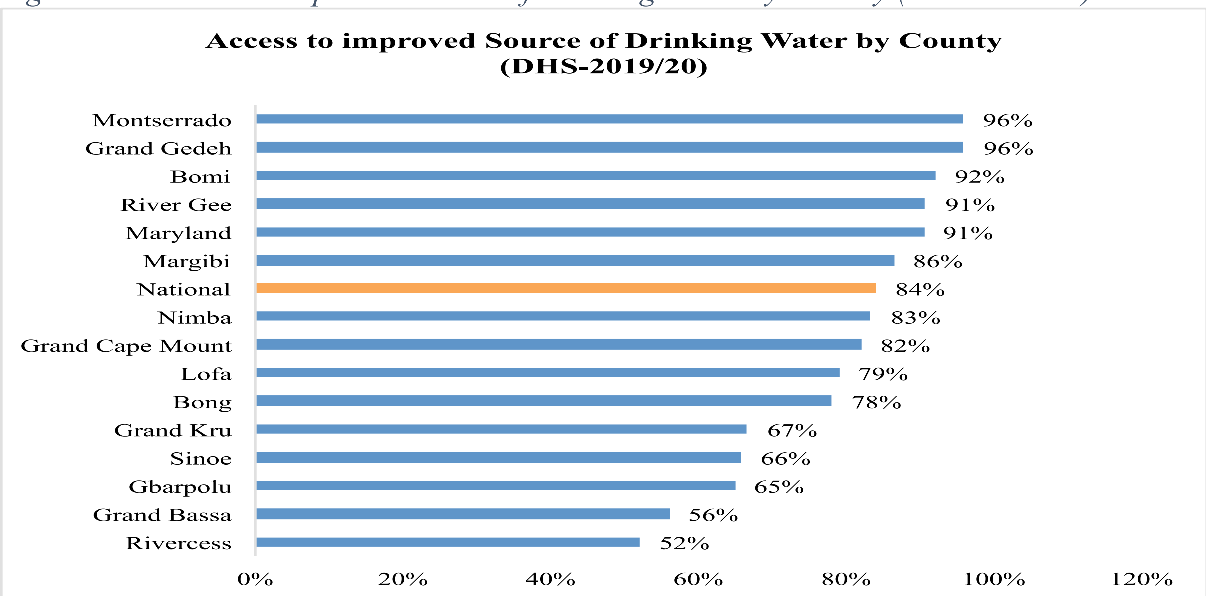
Table 3: Percentage of Recent HIV Testing among Women and Men by County (DHS 2019/20)

Recent HIV Testing among Women and Men by County			
#	*County*	*Women*	*Men*
1	Bomi	30%	18%
2	Bong	25%	21%
3	Gbarpolu	17%	28%
4	Grand Bassa	20%	19%
5	Grand Cape Mount	22%	12%
6	Grand Gedeh	27%	25%
7	Grand Kru	19%	6%
8	Lofa	19%	25%
9	Margibi	23%	29%
10	Maryland	22%	7%
11	Montserrado	21%	23%
12	Nimba	25%	22%
13	Rivercess	20%	12%
14	River Gee	22%	17%
15	Sinoe	15%	13%
	National	***22%***	***32%***

2.6 Environmental Factors

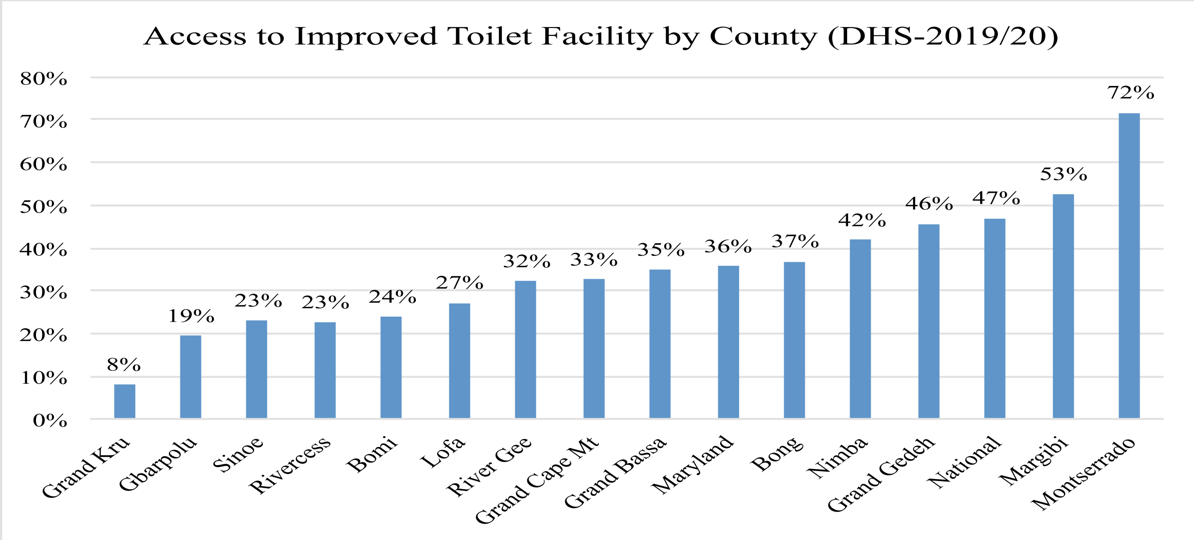
Liberia faces challenges related to vector-borne diseases, inadequate sanitation, and limited access to clean water, contributing to the burden of infectious diseases. Eighty-four (84%) percent of households have access to improved source of drinking water. Figure 13 displays access to improved source of drinking water by county.

Figure 13: Access to Improved Source of Drinking Water by County (DHS 2019/20)



Nearly half (47%) of households in Liberia used improved toilet facilities. However, access to toilet facility is very low in Grand Kru (8%) and Gbarpolu 19%). Figure 14 depicts access to improve source of drinking water.

Figure 14: Access to Improved Toilet Facility by County (DHS 2019/20)



Climate-related events such as flooding and droughts, can disrupt livelihoods and exacerbate health vulnerabilities.

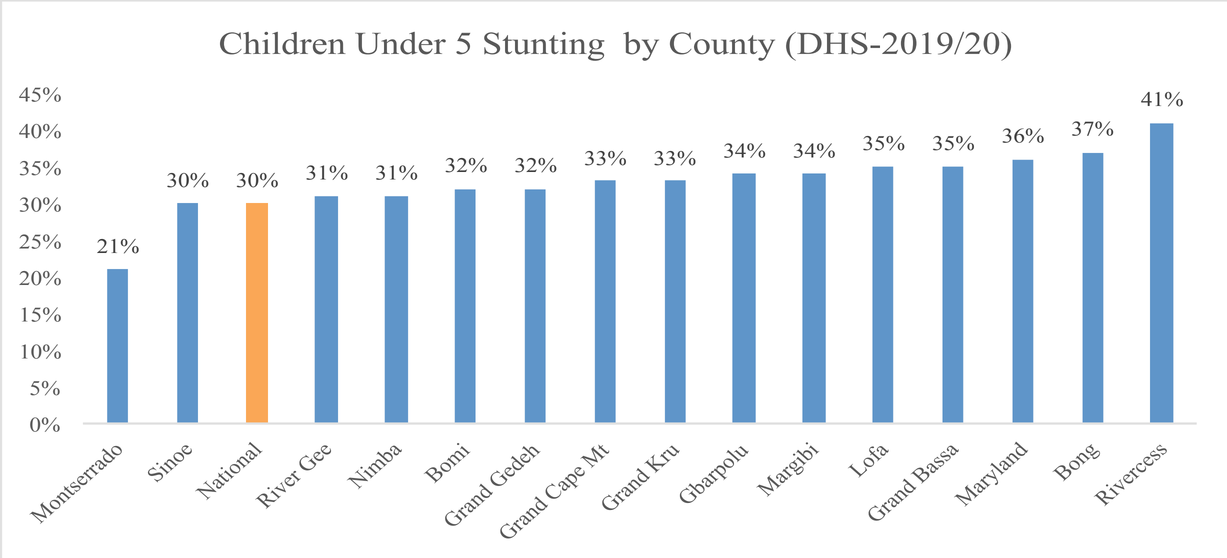
2.7 Access to Nutritious Food

Limited access to nutritious food and reliance on subsistence agriculture can lead to malnutrition and related health issues. According to the most recent STEPS Survey conducted in 2021, the mean number of days fruits are consumed in a typical week in Liberia is two, while the number of days vegetables are consumed in a typical week is 3.2[16].

Thirty percent (30%) of children under age 5 are stunted (short for their age), 3% are wasted (thin for their height), 11% are underweight (thin for their age), and 4% overweight (heavy for their height). Overall, 3% of children age 6-23 months were fed a minimum acceptable diet in the 24 hours before the survey, 5% of women age 15-49 are thin, while 37% are overweight or obese[17]. Stunting is prevalence in Rivercess (41%), and Montserrado has the lowest stunting rate. Figure 15 depicts the percentage of children under age five that are stunted.

[16] Liberia STEPS Survey, 2021
[17] Liberia Demographic and Health Survey, 2019/20

Figure 15: Children Under 5 Stunting by County (DHS 2019/20)

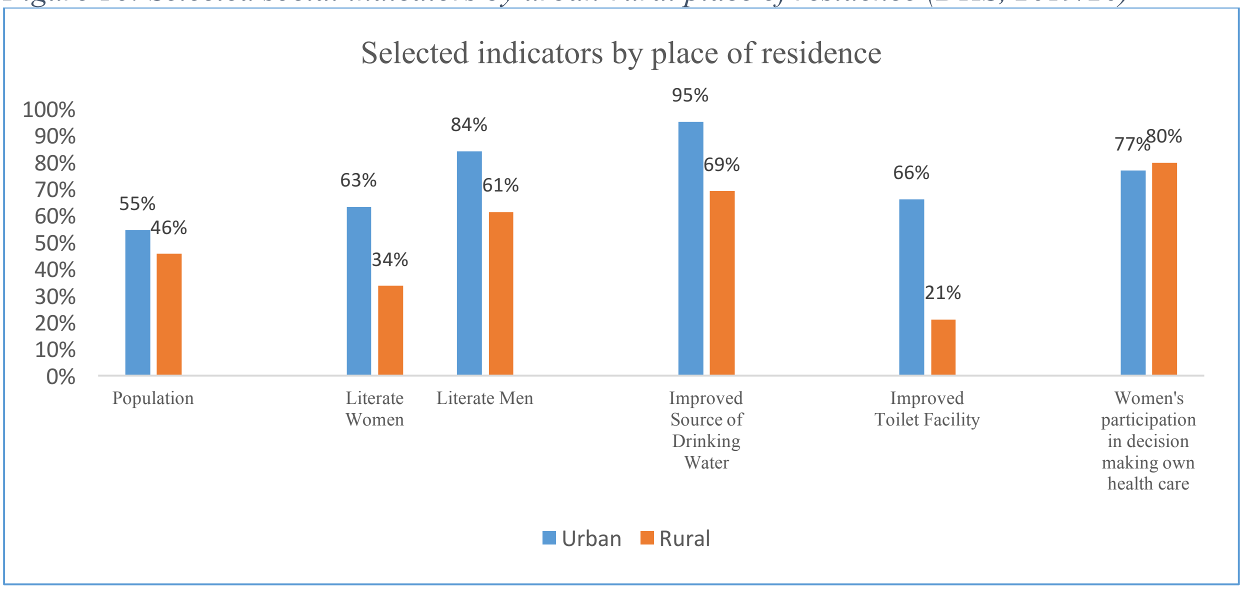


2.8 Rural-Urban Place of Residence

Place of residence, whether urban or rural can significantly impact a person's health in various ways. Urban areas typically have better access to healthcare facilities, including hospitals, clinics, and specialized medical services. Rural areas may have limited healthcare resources, leading to delayed or inadequate care.

Urban areas often face issues like air pollution, noise pollution, and higher population density, which can contribute to health problems such as respiratory issues, stress, and increased disease transmission, while rural areas may have cleaner air and a quieter environment, which can be beneficial for mental health, and overall well-being. Figure 16 displays selected social indicators by urban-rural place of residence.

Figure 16: Selected social indicators by urban-rural place of residence (DHS, 2019/20)



2.9 Housing condition

Housing condition is a critical determinant of health because it directly impacts the physical and mental well-being of individuals and communities. Poor housing conditions can have adverse effects on health. Inadequate housing may expose individuals to various environmental hazards such as mold, lead, asbestos, and pests. These can lead to respiratory problems, allergies, and other health conditions.

Nearly half (49%) of the population lives in homes floor that are constructed by concrete (cement), while 40.7% were made of earth, sand and mud. About a quarter (23.9%) of households have access to electricity.

An outer wall serves as a crucial barrier against weather, intruders, and provides structural support. It provides insulation, security and aesthetic appeal, contributing significantly to the overall durability and safety of a dwelling place. In Liberia, only 47% of homes outer walls are made of durable materials such as concrete, cement blocks and clay bricks. Over half (53%) of dwelling places are made of temporary materials such as mud & sticks, wood, zinc, among others. Table 4 below shows households dwelling places by materials used for the outer walls.

Table 4: Households dwelling places by materials used for outer walls by county, 2022 (Census)

County	Stone Concrete	Cement Blocks	Clay Bricks	Zinc or Iron	Wood or Board	Mud & Bricks	Mud & Stick	Reed Bamboo Grass or	Other
Bomi	6.6	16.8	7.4	1.3	0.8	16.2	49.8	1.1	0.0
Bong	6.9	11.7	7.7	0.9	0.5	28.2	43.9	0.1	0.1
Gbarpolu	6.4	2.5	2.5	1.3	0.6	17.9	68.2	0.6	0.0
Grand Bassa	6.2	22.2	3.3	2.4	0.9	11.8	51.8	1.3	0.2
Grand Cape Mount	7.5	17.8	8.1	3.3	1.4	24.6	35.2	2.0	0.1
Grand Gedeh	6.1	9.9	4.7	1.2	1.0	19.8	56.3	0.8	0.3
Grand Kru	4.5	4.4	1.1	0.6	1.4	7.5	79.2	1.3	0.1
Lofa	5.9	7.5	5.3	0.8	1.4	63.7	15.0	0.3	0.0
Margibi	8.4	35.2	12.8	7.1	0.8	13.7	21.0	0.8	0.1
Maryland	6.8	11.0	1.2	1.5	1.0	8.0	70.1	0.2	0.1
Montserrado	14.9	60.8	2.8	11.5	0.3	5.2	4.1	0.1	0.2
Nimba	8.0	11.7	11.4	0.9	0.8	51.3	15.5	0.2	0.2
River Cess	2.8	3.7	1.1	0.7	1.0	9.4	79.7	1.2	0.6
River Gee	5.3	2.6	0.8	0.2	0.5	8.2	81.6	0.7	0.1
Sinoe	7.6	11.5	2.5	1.9	2.6	10.8	62.1	0.6	0.3
Total	***9.9***	***31.8***	***5.2***	***5.5***	***0.7***	***19.2***	***27.1***	***0.5***	***0.1***

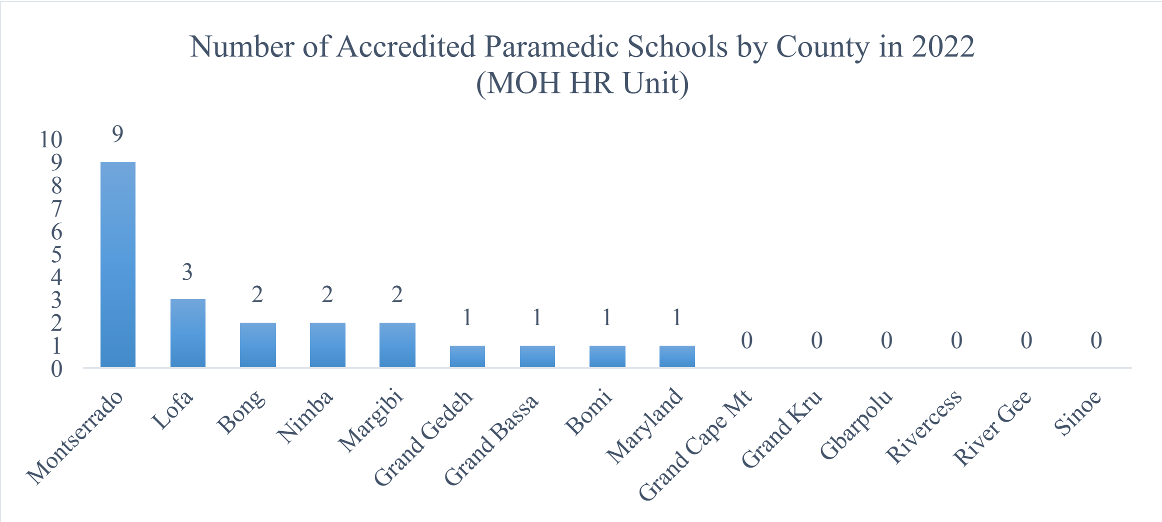
Addressing social determinants of health in Liberia requires multi-sectoral efforts, including investments in healthcare infrastructure, education, poverty reduction, gender equality, and environmental sustainability. Additionally, efforts to promote good governance and reduce corruption are essential for improving health outcomes and overall well-being in Liberia.

2.10 Health Training Institutions

Health training institutions are educational facilities or organizations that provide training and education in various aspects of healthcare and medicine. These institutions play a crucial role in preparing individuals for careers in healthcare professions. They can include medical schools, nursing schools, pharmacy schools, and various vocational or technical schools that offer programs related to healthcare, such as Laboratory Aid, bio-technology, dispensary, among others. Health training institutions are essential for producing a skilled and qualified workforce to meet the healthcare needs of communities and populations.

As of 2022, Liberia has one medical school, one school of pharmacy, five accredited Laboratory training institutions, and dozens of nursing and midwifery schools. There are six counties (Gbarpolu, Grand Cape Mt, Grand Kru, Rivercess, River Gee, and Sinoe) with no healthcare training institution. Figure 17 presents the number of accredited paramedic schools by county in 2022. Annex B provides a list of paramedic training institutions in Liberia.

Figure 17: Number of Accredited Paramedic Schools by County in 2022 (MOH-HR Unit)



3.0 Health Inequality and Equity

3.1 Introduction

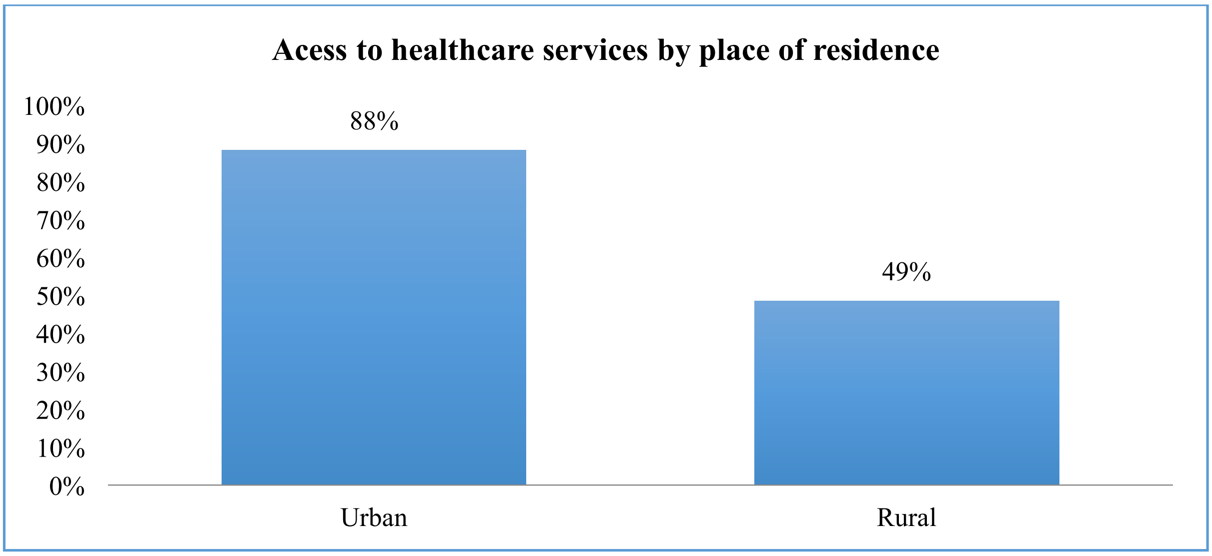
Health equity refers to the concept of everyone having a fair opportunity to attain their highest level of health, regardless of social, economic, or demographic factors. It implies that all individuals should have access to quality healthcare and the conditions necessary for good health, without discrimination or barriers.

Health inequality, on the other hand, pertains to disparities in health outcomes and access to healthcare services among different populations. These disparities often stem from socioeconomic factors, ethnicity, gender, geography, and more. Health inequality can lead to certain groups having poorer health outcomes and reduced access to healthcare compared to others.

3.2 Access to healthcare services

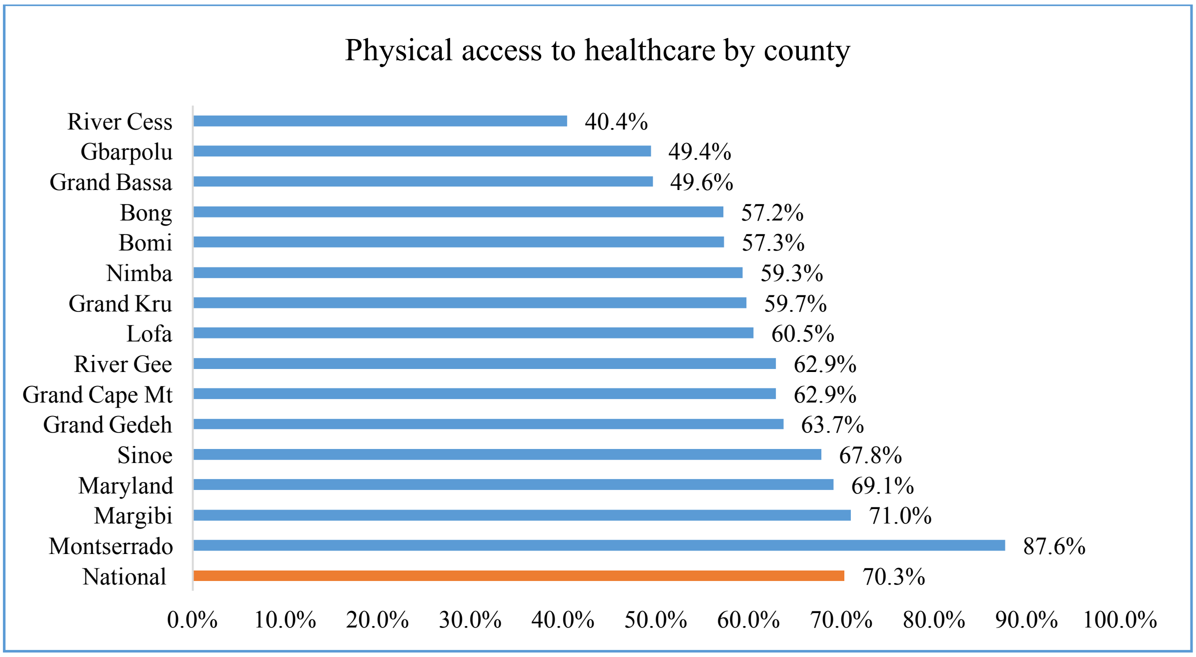
Access to health care is fundamental to growth and development and the attainment of universal health coverage in Liberia. Access to health care is higher in urban areas than rural. As depicted in figure 18, eighty-eight (88%) percent of urban dwellers have access to healthcare compared to 49% of rural dwellers (2022 Population Census).

Figure 18: Access to health services by place of residence



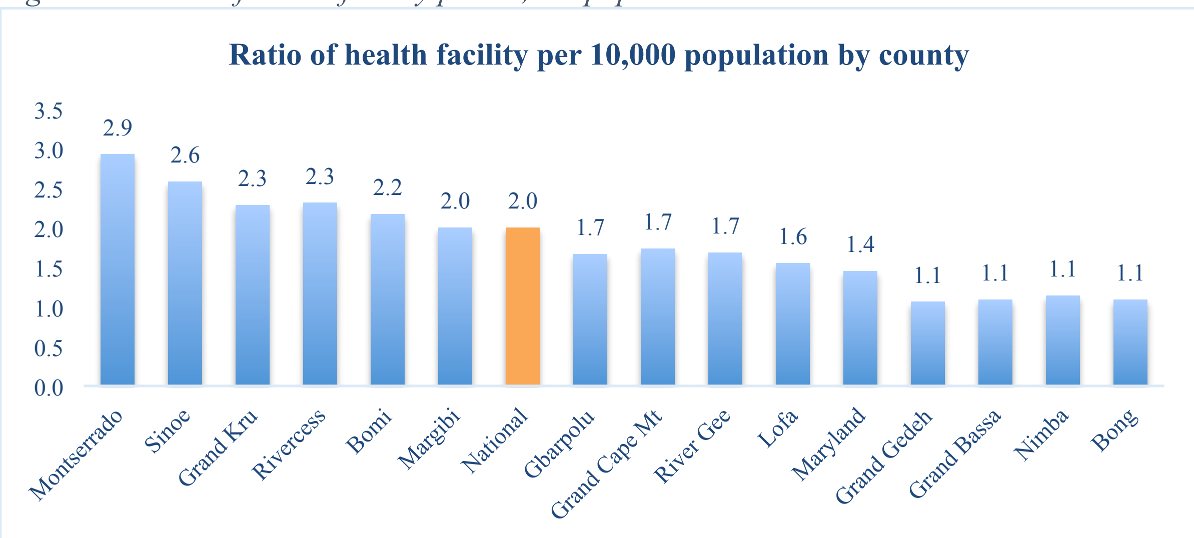
Nationally, 70.3% of inhabitants of Liberia have access to healthcare. Counties with less than 50% of their population having access to health care are Rivercess (40.4%), Gbarpolu (49.4%) and Grand Bassa. However, Montserrado has the highest access of 87.6%, followed by Margibi (71%). Figure 19 displays access to healthcare by county.

Figure 19: Physical access to healthcare by county



The ratio of health facility per 10,000 population is an indicator that measures physical access to health care or health facility density. It refers to the number of persons that are served by a health facility (both public and private) in which general health services are offer within a county. In Liberia, two health facilities are serving 10,000 people. However, one-third of the counties (5) have two or more health facilities serving 10,000 people. Figure 20 shows the health facility density in Liberia.

Figure 20: Ratio of health facility per 10,000 population



This standardized indicator measures levels of access to health services by the designated populations. It can be used to identify under-served areas and will allow comparisons within and between counties. Geographic mapping will allow identification of where there are coverage gaps for certain populations. This indicator can contribute to monitoring progress on achieving universal health coverage (SDG target 3.8); reducing child mortality (SDG target 3.2); improving maternal health (SDG target 3.1); and combating HIV/AIDS (SDG target 3.3).

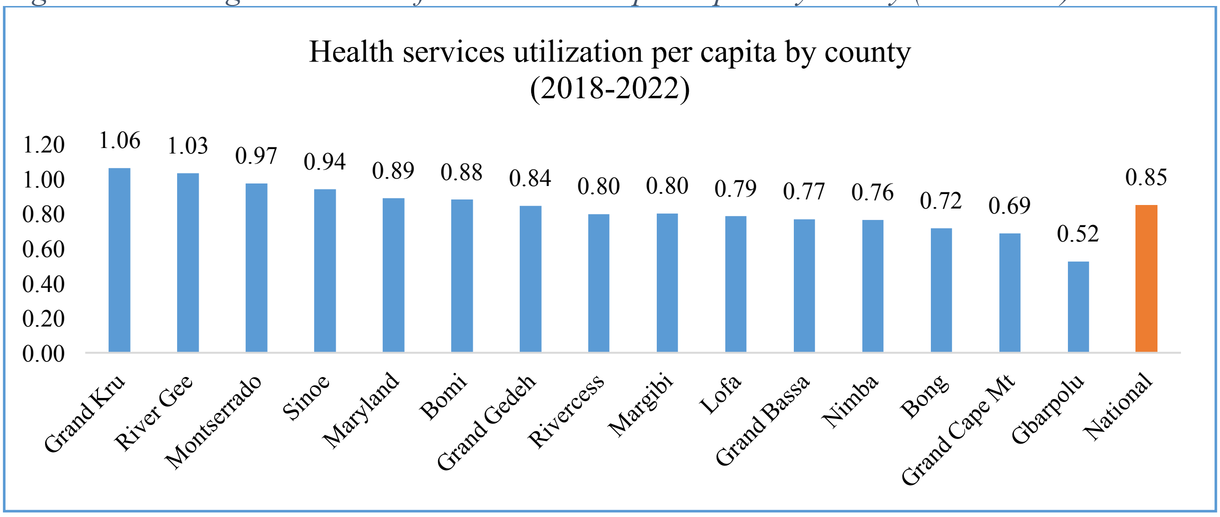
3.3 Health services utilization

Health services utilization in Liberia, like in many low-income countries, has historically faced challenges. Several factors contribute to this situation:

a) Inadequate healthcare infrastructure and limited access to healthcare facilities, especially in rural areas have been barriers to healthcare utilization.
b) High poverty rates and limited health insurance coverage have made it difficult for many Liberians to afford healthcare service.
c) Perceptions of low-quality care at public healthcare facilities have discouraged some individuals from seeking medical attention.
d) Traditional healing practices are still prevalent in some regions, and this can deter people from seeking modern healthcare services.
e) Limited health education and awareness in some communities can lead to misconceptions about healthcare services and disease prevention.

Health services utilization is very low in Liberia. Over the past 5years (2018-2022), utilization has been below one visit per capita which is 83% reduction of the WHO recommended five visits per capita per year. The highest service utilization was found in Grand Kru (1.06 per capita), River Gee (1.03 per capita) and Montserrado (0.97 per capita) counties. The lowest service utilization was reported from Gbarpolu (0.52 per capita), Grand Cape Mt (0.69 per capita) and Bong (0.72 per capita) counties[18]. Figure 21 displays the average utilization of health services per capita by county from 2018-2022.

Figure 21: Average utilization of health services per capita by county (2018-2022)



Efforts have been made to improve health services utilization in Liberia, including investments in healthcare infrastructure, health education campaigns, and initiatives to increase access and affordability. Additionally, the Liberia Health Equity Fund (LHEF) has been established to improve health financing and accessibility for the poorest Liberians.

[18] MOH Health Management Information System (DHIS2)

3.4 Coverage of health services

Health service coverage refers to the extent to which a population has access to essential healthcare services, which are necessary for promoting, maintaining, and improving their health. Comprehensive health service coverage encompasses a range of healthcare components, including:

a) Access to basic healthcare services such as routine check-ups, vaccinations, maternal and child health services, and treatment for common illnesses.

b) Availability of preventive measures like immunization, cancer screenings, and health education to avoid illnesses.

c) Comprehensive care for expectant mothers, including prenatal care, as well as pediatric healthcare services for children.

d) Access to specialized medical services like cardiology, oncology, and surgery for more complex conditions.

e) Prompt access to emergency medical care for accidents, trauma, and life-threatening situations.

f) Access to mental health assessments, counseling, and treatment for individuals dealing with mental health disorders.

Universal health coverage (UHC) is a global goal that aims to ensure all individuals and communities have access to necessary health services without suffering financial hardship. Achieving UHC requires addressing various factors, including healthcare infrastructure, affordability, and healthcare workforce availability. Government and international organizations work to expand healthcare coverage by implementing policies, improving healthcare delivery systems, and reducing financial barriers through insurance schemes and subsidies.

Efforts to enhance health service coverage are essential for promoting public health, reducing health disparities, and ultimately improving the quality of life for the population.

3.4.1 Childhood Immunization

Childhood vaccinations in Liberia typically include immunizations against diseases such as:

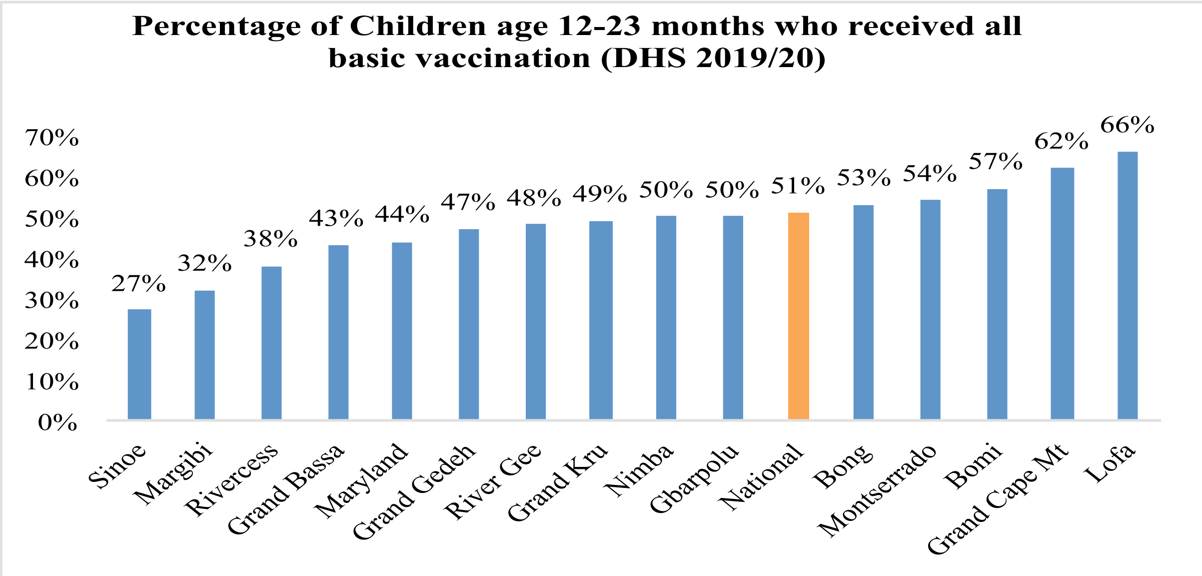
(a) Measles
(b) Polio
(c) Tuberculosis (BCG)
(d) Diphtheria
(e) Tetanus
(f) Pertussis (Whooping Cough)
(g) Hepatitis B
(h) Haemophilus influenzae type b (Hib)
(i) Pneumococcal disease
(j) Rotavirus-Severe Gastroenteritis

(k) Meningitis
(l) Yellow Fever
(m) Typhoid Fever

These vaccines are essential to protect children from preventable diseases and are often administered according to the recommended schedule set by the Liberian Ministry of Health and international health organizations like the World Health Organization (WHO) and UNICEF.

In Liberia, the proportion of children 12-23months that have received all basic vaccines varies from county to county, with Sinoe having little over a quarter (27%) of their children receiving all basic vaccines. On the other hand, Lofa has two-thirds of the children (66%) receiving all basic vaccinations. Figure 22 depicts the percentage of children 12-23 months who received all basic vaccines by county.

Figure 22: Percentage of Children 12-23 months who received all basic vaccinations by county

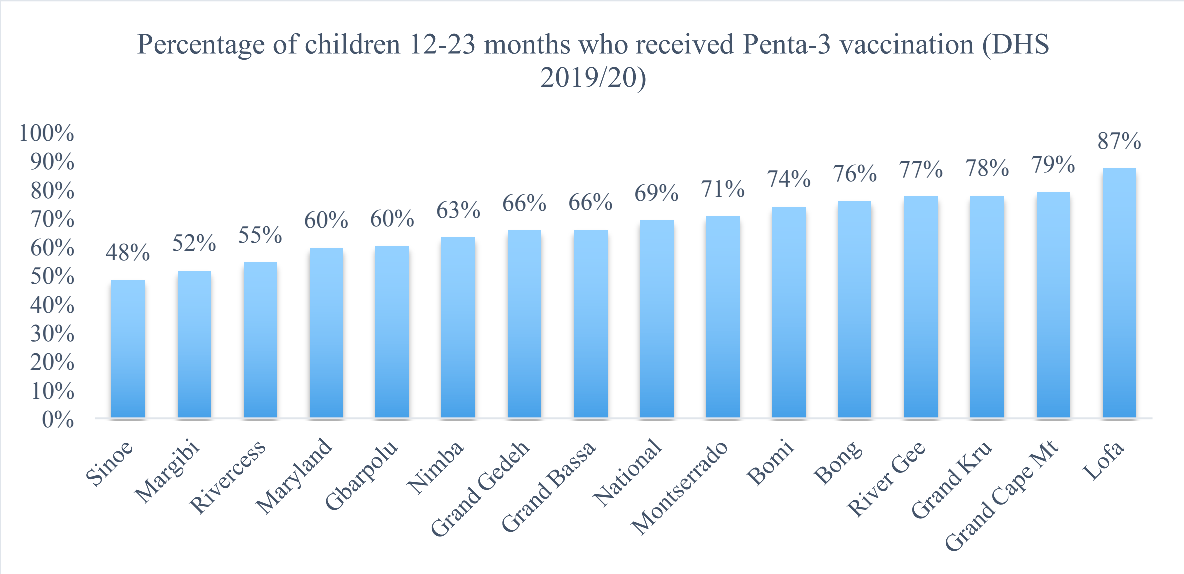


Pentavalent Vaccine Coverage

Pentavalent vaccination is a combination vaccine that provides protection against five different diseases. It typically includes vaccines for diphtheria, tetanus, pertussis (whooping cough), hepatitis B and haemophilus influenza type b (Hib). This combination vaccine is often administered to infants to increase their protection against multiple diseases.

Penta-3 coverage is 69% in Liberia, and coverage varies across counties. Sinoe (48%), Margibi (52%) and Rivercess (55%) counties recorded the lowest penta-3 coverage while Lofa (87%), Grand Cape Mt (79%), and Grand Kru (78%) reported the highest penta-3 coverage. Figure 23 depicts the percentage of children 12-23 months who received penta-3 vaccination.

Figure 23: Percentage of children 12-23 months who received Penta-3 vaccination (DHS 2019/20)

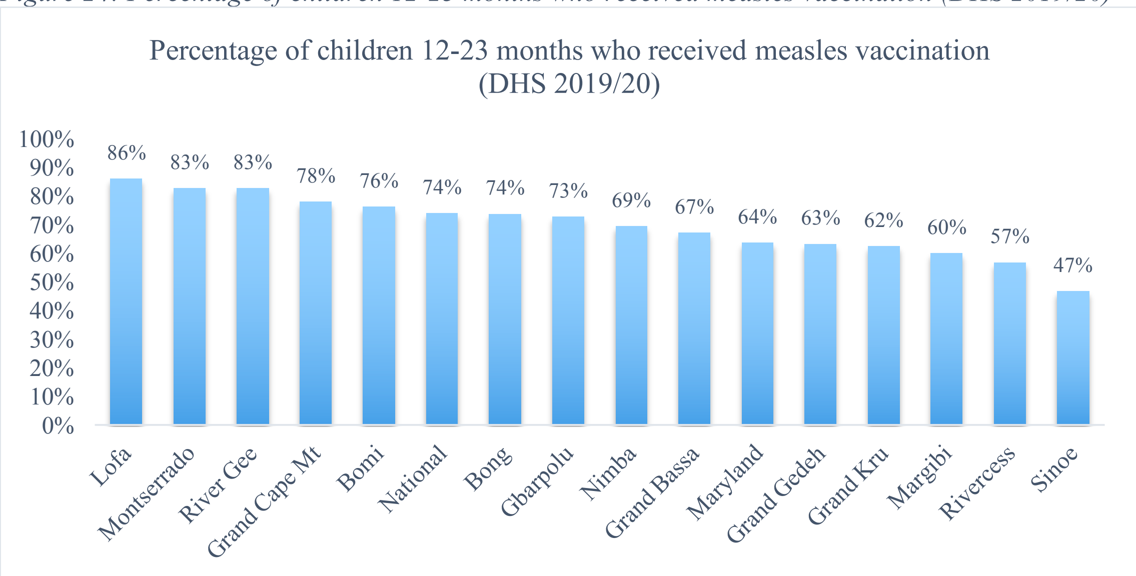


Measles Vaccine Coverage

Measles is a highly contagious viral disease caused by the measles. It typically presents with symptoms like high fever, cough, runny nose, and a red rash. Vaccination with the measles, mumps, and rubella (MMR) vaccine is highly effective in preventing measles.

Measles coverage is 74% nationally, with disparities across counties. The lowest coverage was found in Sinoe (47%) and Rivercess (57%) counties while Lofa (86%), Montserrado (83%), and River Gee (83%) counties reported the highest coverage. Figure 24 percentage of children 12-23 months who received measles vaccination.

Figure 24: Percentage of children 12-23 months who received measles vaccination (DHS 2019/20)

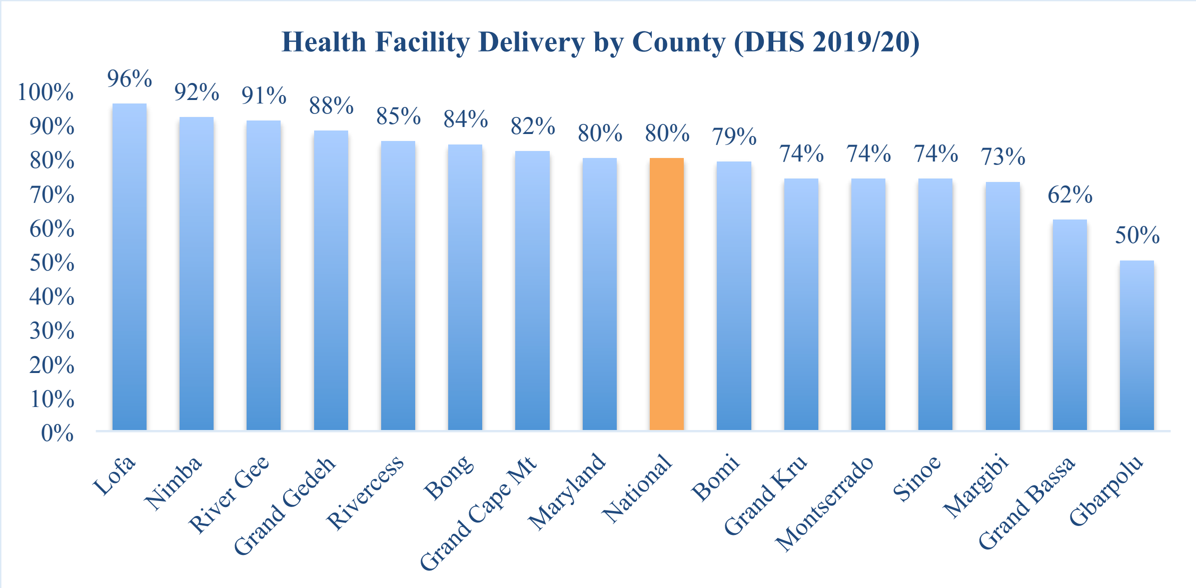


3.4.2 Health facility delivery

Health facility delivery refers to the practice of giving birth in a healthcare facility such as hospital or clinic, under the care of trained medical professionals. It is generally considered a safe option compared to home births, as it allows for access to medical interventions and emergency care if complications arise during childbirth. Health facility delivery is recommended in many cases to ensure the well-being of both the mother and the baby. Health services are presumed free especially for pregnant women and children under age five, to reduce maternal mortality. However, there are rumors that poor mothers are detained at health facilities for lack of money. Unfortunately, they are punished contrary to the principle of achieving UHC.

In Liberia, majority of the deliveries occur in healthcare facilities. However, there are variations across counties. The proportion of health facility delivery is highest in Lofa (96%), Nimba (92%), and River Gee (91%) Counties, while Gbarpolu (50%) and Grand Bassa (62%) Counties reported the lowest proportion of health facility delivery. Figure 25 shows the percentage of health facility delivery by county.

Figure 25: Health facility delivery by county



Skilled birth attendants are healthcare professionals trained to provide care during childbirth. They play a crucial role in ensuring safe deliveries. Here are some key aspects of services provided by skilled birth attendants:

a) Skilled birth attendants provide prenatal care to monitor the health of the mother and the fetus throughout pregnancy.

b) They assist the mother during labor, ensuring comfort, and monitoring the progress of labor.

c) Skilled birth attendants are trained to manage the actual delivery process, including guiding the mother through the stages of labor and assisting with the birth of the baby.

d) In case of complications or emergencies, they are equipped to provide immediate medical care to both the mother and the baby. They employ Basic Life Saving Skills (BLSS).

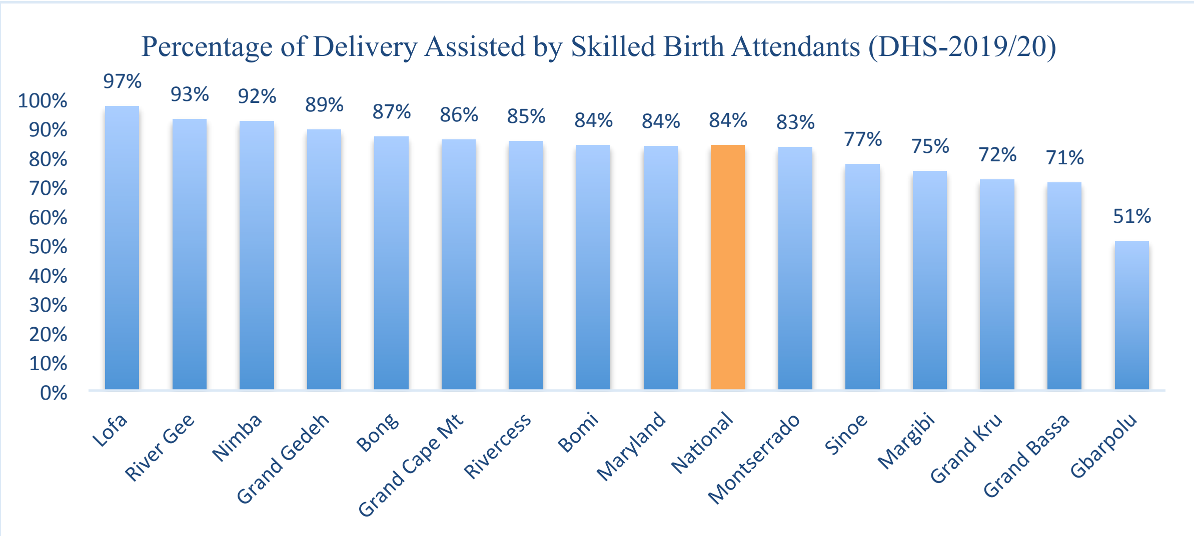
e) Skilled birth attendants continue to care for the mother and baby after delivery, ensuring their well-being and addressing any postpartum issues.

f) They offer guidance to the mother on topics such as breastfeeding, newborn care, and family planning.

The presence of skilled birth attendants significantly reduces the risks associated with childbirth and contributes to healthier outcomes for both the mother and the newborn.

Delivery assisted by skilled birth attendance is encouraging in Liberia. Lofa County (97%) has the highest proportion (97%) of delivery assisted by skilled birth attendants, while Gbarpolu County (51%) reported the lowest delivery assisted by skilled birth attendants. Figure 26 depicts the percentage of delivery assisted by skilled birth attendants.

Figure 26: Percentage of delivery assisted by skilled birth attendants



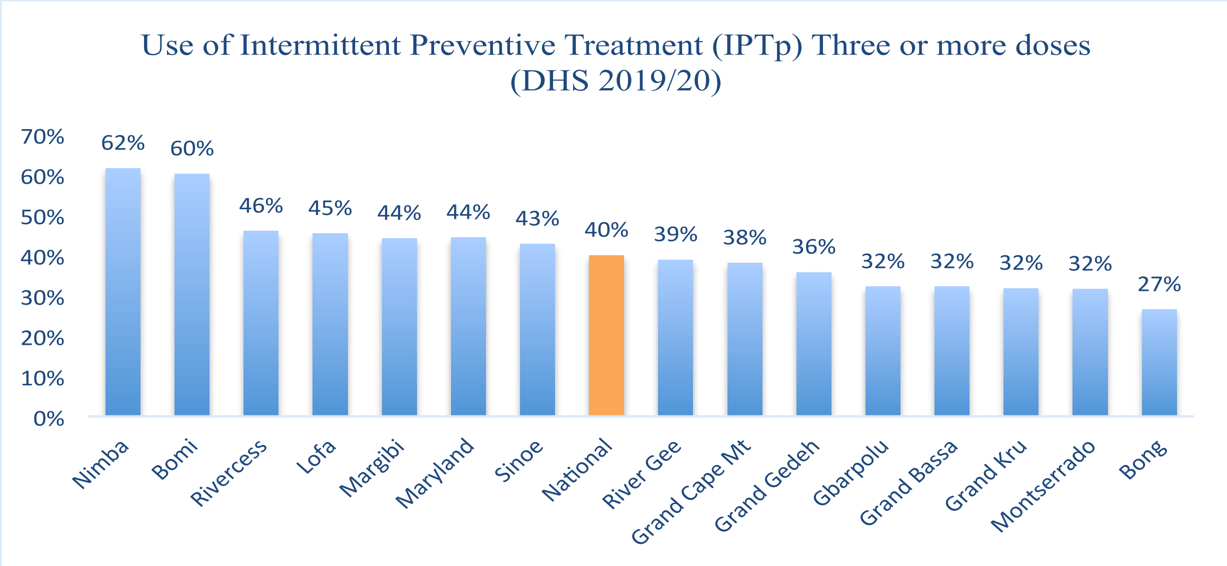
3.4.3 Malaria prevention

Intermittent preventive treatment (IPT) is a medical strategy primarily used in the context of preventing and controlling certain infectious diseases, most notably malaria. It involves the administration of preventive medication at specific intervals to individuals at risk of contracting the disease.

In summary, intermittent preventive treatment is a vital tool in the fight against infectious diseases, particularly malaria. It involves administering preventive medication at specific intervals to at risk population, ultimately reducing the disease's impact and saving lives.

The administration of IPTp is low in Liberia due partly to erratic stock out of SP/Fansidar at health facilities. Nimba County has the highest administration (62%) of three or more doses of IPT to pregnant women, while Bong County administered (27%) the lowest doses. Figure 27 presents the use of IPT3 or more doses by county.

Figure 27: Administration of IPTp 3 or more doses by county

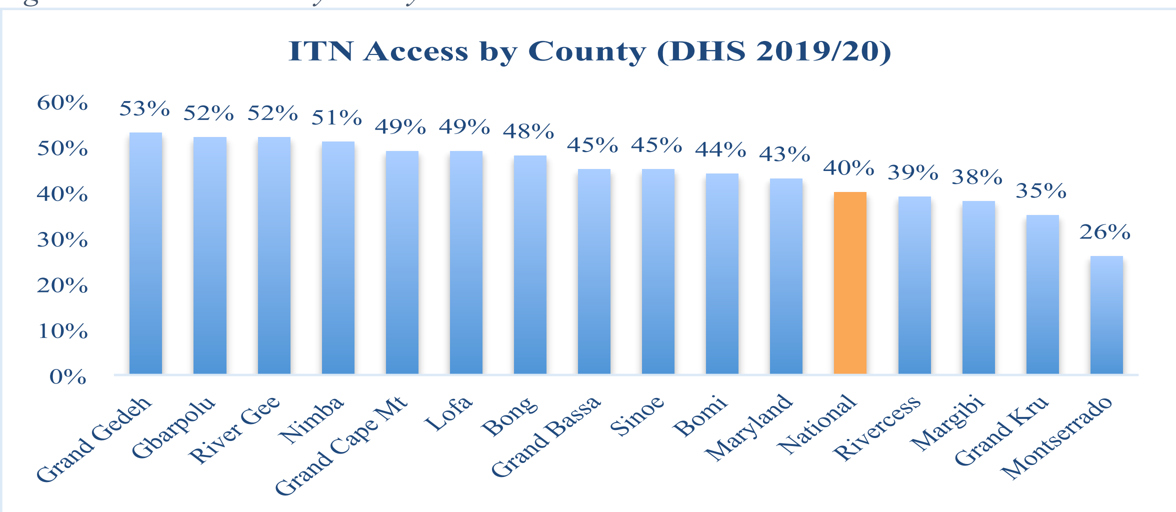


Access to insecticide-treated Nets (ITNs) is a critical component of global efforts to combat vector-borne diseases, particularly malaria. These bed-nets are a cost-effective and proven method for preventing the transmission of malaria and other mosquito-borne illnesses.

Access to ITNs is a fundamental part of global health efforts to reduce the burden of vector-borne diseases, and continued commitment to improving this access is essential for achieving meaningful progress in disease prevention and control.

Access to ITN is low in Liberia and the proportion of households with ITNs vary from county to county. Grand Gedeh County (53%) has the highest access to ITN, followed by Gbarpolu County (52%), and River Gee County (52%). Also, Montserrado County (26%), and Grand Kru County (35%) have the lowest access to ITN. Figure 28 shows ITN access by county.

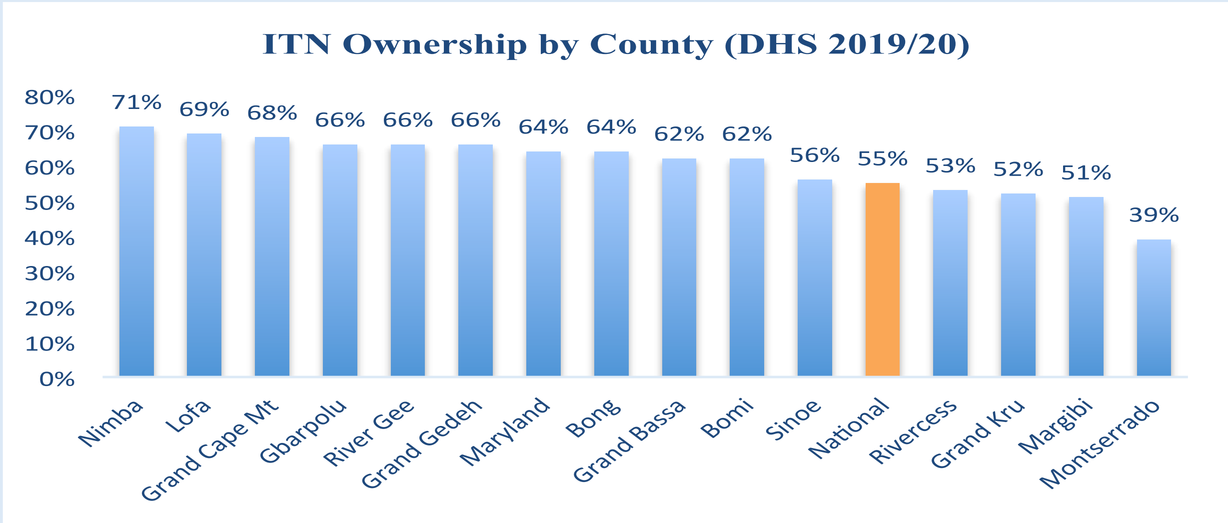
Figure 28: ITN access by county



ITN ownership varies from county to county, with Nimba County households (71%) having the highest ITN ownership, followed by Lofa County (59%), and Grand Cape Mount County

(68%). The county with the lowest ITN ownership is Montserrado County (39%). Figure 29 presents ITN ownership by county.

Figure 29: ITN ownership by county



3.5 Health status (morbidity, mortality)

Liberia's health status faced various challenges but also showed signs of improvement in some areas. Liberia has struggled with infectious diseases, including outbreaks of diseases like Ebola Virus Disease (EVD), Lassa fever and Corona Virus Disease (COVID-19). The EVD outbreak, which peaked in 2014-2015, and the COVID-19 pandemic had a devastating impact on the country's healthcare system.

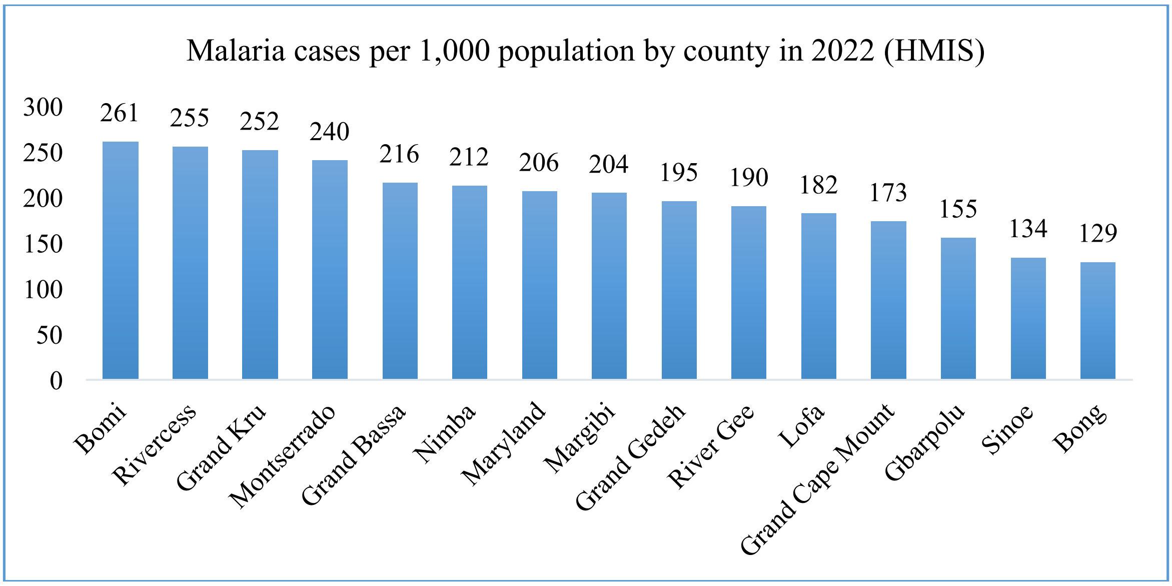
Liberia's healthcare infrastructure is limited and faced significant challenges. Many health facilities, particularly in rural areas, need of improvement in terms of equipment, staffing, and access to essential services.

Maternal and child health indicators are of great concern. High maternal mortality rates of 742 deaths per 100,000 live births and child malnutrition are among the challenges the country faced. Access to skilled birth attendants and healthcare for pregnant women and children is a priority.

Malaria is a major public health concern in Liberia. Efforts have been made to control and prevent malaria through initiatives such as the distribution of bed nets, malaria testing and treatment.

Malaria is endemic in Liberia and accounts for the highest attendance of out-patient department. In 2022, Bomi (261), Rivercess (255) and Grand Kru (252) reported the highest incidence rate per 1,000 population, while Bong (129) and Sinoe (134) Counties recorded the lowest incidence of malaria. Figure 30 shows the number of malaria cases per 1,000 population by county in 2022.

Figure 30: Malaria cases per 1,000 population by county in 2022 (HMIS)



Liberia has made progress in immunization programs, aiming to protect children against vaccine-preventable diseases. The country experienced outbreaks of vaccine preventable diseases, as a result of low immunization coverage and high cohort of unimmunized children. Access to healthcare services and healthcare financing remained significant issues, with many people lacking the financial means to cover healthcare costs.

It's important to note that the Liberian government and international organizations have been working to address these health challenges through various healthcare programs and initiatives.

3.5.1 Malnutrition

Stunting

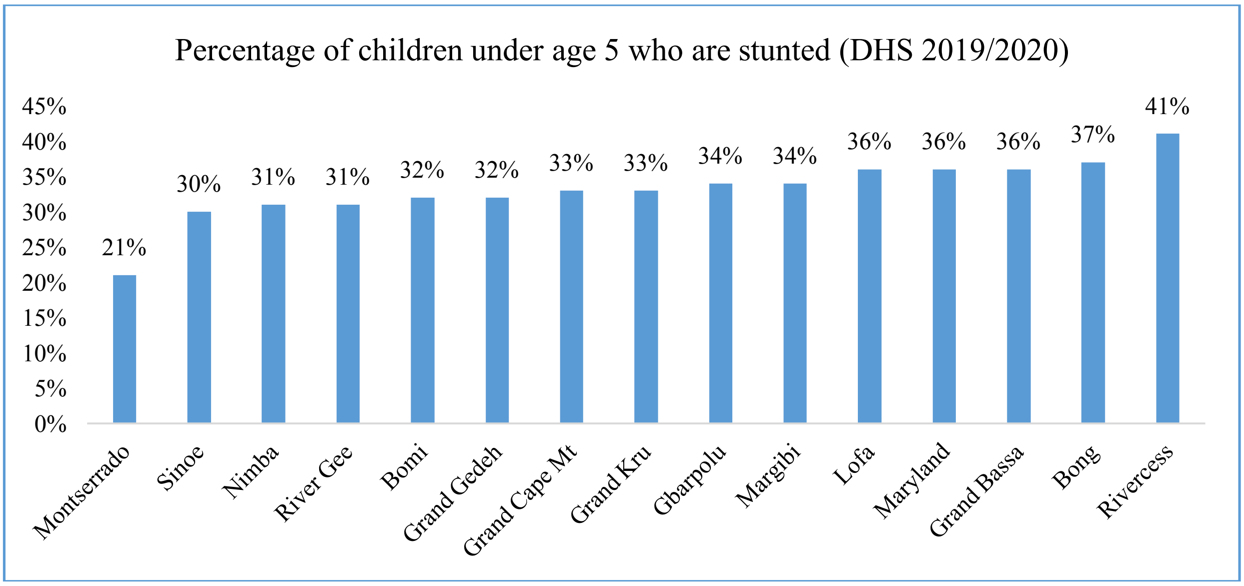
Stunting in children under five is a significant public health concern. It is also known as "chronic malnutrition", refers to the impaired growth and development of a child due to prolonged nutritional deficiencies, recurrent infections, and inadequate care. It is typically measured by comparing a child's height-for-age to established growth standards. Key points about stunting in children under five include:

a) Stunting can have long-term and irreversible effects on a child's physical and cognitive development. It can lead to reduced adult height, diminished cognitive abilities, and increased susceptibility to chronic diseases later in life.
b) Stunting is primarily caused by lack of adequate nutrition, particularly during the critical "1,000-day window" from pregnancy through the first two years of life. Poor maternal nutrition, lack of access to nutrition, lack of access to nutritious food, and unsanitary conditions contribute to stunting.
c) Stunting not only affects individual children but also has broader societal implications. It can hinder a nation's economic development by reducing the productivity of its workforce.

d) To combat stunting, a multi-pronged approach is needed. This includes improving maternal nutrition, promoting breastfeeding, ensuring access to nutritious foods, enhancing sanitation and hygiene, and providing proper healthcare.

Stunting is prevalent in Liberia, with 30% of children under age 5 stunted. However, its severity varies from one county to another with Montserrado County experiencing the lowest rate (21%) of stunting, followed by Sinoe County (30%). The highest rate of stunting is found in Rivercess County (41%) and Bong County (37%)[19]. Figure 31 presents the percentage of children under age 5 who are stunted.

Figure 31: Percentage of children under age 5 who are stunted



Efforts to reduce stunting in children under five are essential to ensure that every child can reach their full physical and cognitive potential. It requires a combination of nutrition, healthcare, and education initiatives, as well as broader socio-economic improvements in affected communities.

Severe Malnutrition

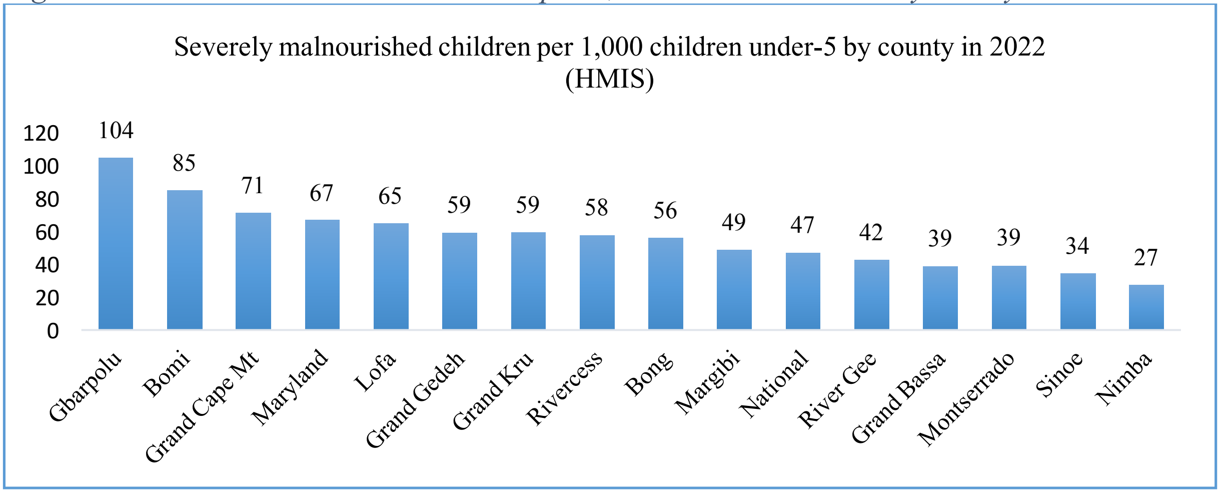
Malnutrition is a condition that results from imbalance between the nutrients a person's body needs and the nutrients they consume. It can manifest as under-nutrition (insufficient intake of essential nutrients) or over-nutrition (excessive intake of certain nutrients), leading to various health problems. Malnutrition can have serious health consequences and affects people of all ages. It's important to address and prevent malnutrition through proper nutrition, health education and healthcare.

Severe malnutrition is a critical and life-threatening form of malnutrition. It occurs when an individual's intake of essential nutrients, such as calories, protein, vitamins, and minerals is significantly inadequate for an extended period. This condition can lead to severe health complications, including weight loss, muscle wasting, weakened immune system, and organ damage. Severe malnutrition often requires immediate medical attention and nutritional rehabilitation to restore the individual's health and prevent further complications.

[19] Liberia Demographic and Health Survey, 2019/20

In Liberia, severe malnutrition is becoming widely spread especially among children under age 5. In 2022, 4.7% of children under age 5 were admitted for severe malnutrition across Liberia. The highest cases of malnutrition were reported from Gbarpolu (104 children per 1,000 under-5), Bomi (85 children per 1,000 under 5), and Grand Cape Mt (71 children per 1,000 under-5). The lowest prevalence of severe malnutrition was found in Nimba (27) and Sinoe (34)[20]. Annex C provides the number of admissions, relapses, and deaths. Figure 32 presents the number of children under age 5 per 1,000 under five by county in 2022.

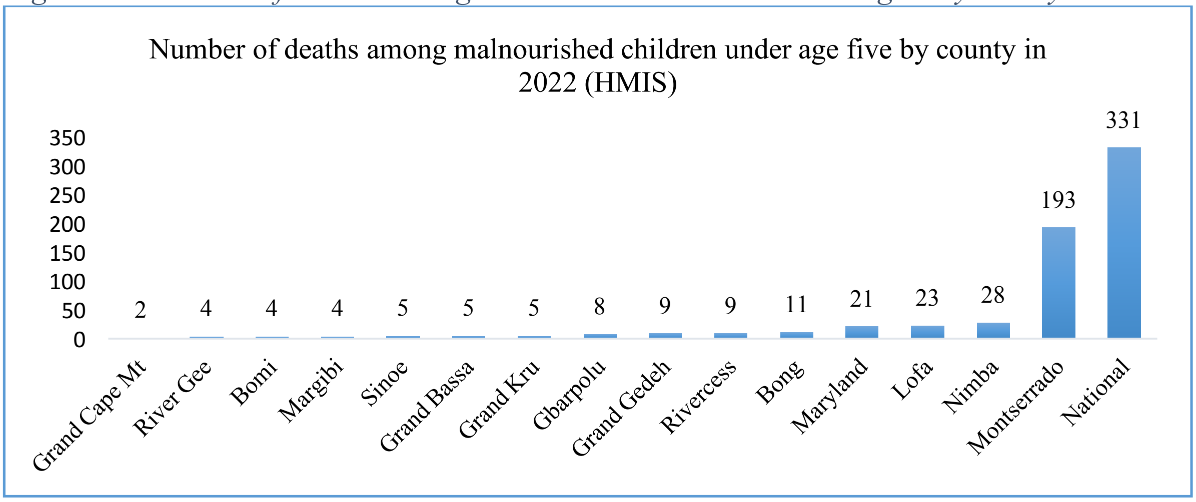
Figure 32: Severe malnourished children per 1,000 children under-5 by county in 2022



Mortality

A total of 331 children under age 5 died of malnutrition in 2022. The number of deaths varies from county to county. Figure 33 shows the number of deaths among malnourished children under age five in 2022.

Figure 33: Number of deaths among malnourished children under age 5 by county in 2022



3.5.2 Tuberculosis (TB)

Tuberculosis (TB) is a contagious bacterial infection that primarily affects the lungs but can also target other parts of the body. It is caused by mycobacterium tuberculosis, a bacterium that

[20] MOH HMIS Data 2022

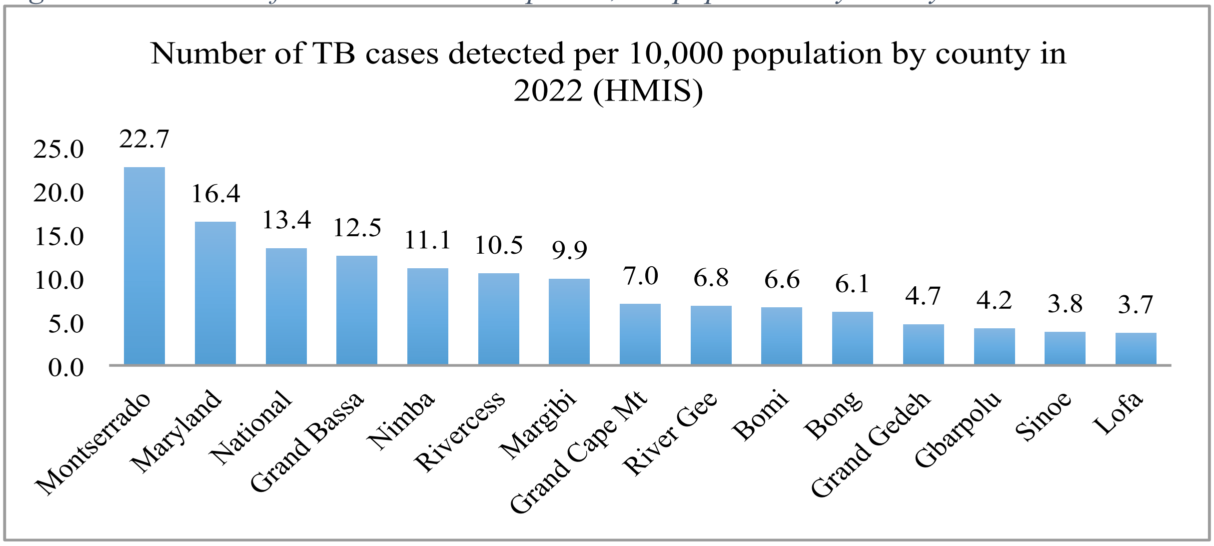
spreads through the air when an infected person coughs or sneezes. TB is primarily spread through the inhalation of airborne droplets containing the bacteria. Close and prolonged contact with an infected person leads to transmission.

Common symptoms of TB include persistent cough, fever, weight loss, night sweats, and fatigue. In more severe cases, it can lead to chest pain, coughing up blood, and difficulty breathing. TB is treatable with a combination of antibiotics over several months. Completing the full course of treatment is crucial to prevent drug resistance.

TB Cases Detected

Tuberculosis case detection is the process of identifying individuals who have active TB infections. This is important for controlling the spread of the disease and providing appropriate treatment. Early and accurate case detection is crucial for timely treatment and preventing the further spread of TB. The number of TB cases detected per 10,000 population in 2022 was 13.4 cases. The highest detection rate was found in Montserrado (22.7 cases per 10,000 people), Maryland (16.4 per 10,000 people), and Grand Bassa (12.5 cases per 10,000 people). Lofa County reported a low number of TB cases of 3.7 per 10,000 people. Figure 34 depicts the number of TB cases detected in 2022 by county.

Figure 34: Number of TB cases detected per 10,000 population by county in 2022

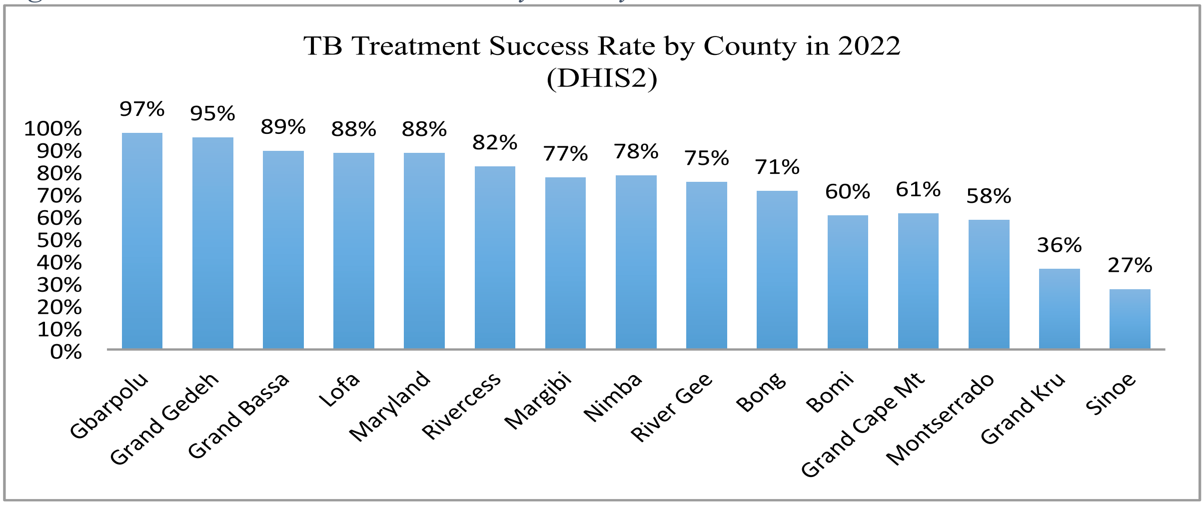


TB Treatment Success Rate

The treatment success rate for TB varies depending on several factors, including the type of TB, the patient's adherence to treatment, and the healthcare infrastructure in a given county or geographic location.

In 2022, Gbarpolu County recorded the highest TB treatment success rate of 97%, followed by Grand Gedeh County (95%) and Grand Bassa County (89%). The lowest treatment success rate was reported from Sinoe (27%) and Grand Kru (36%). Figure 35 presents the TB treatment success rate by county in 2022.

Figure 35: TB Treatment Success Rate by County in 2022

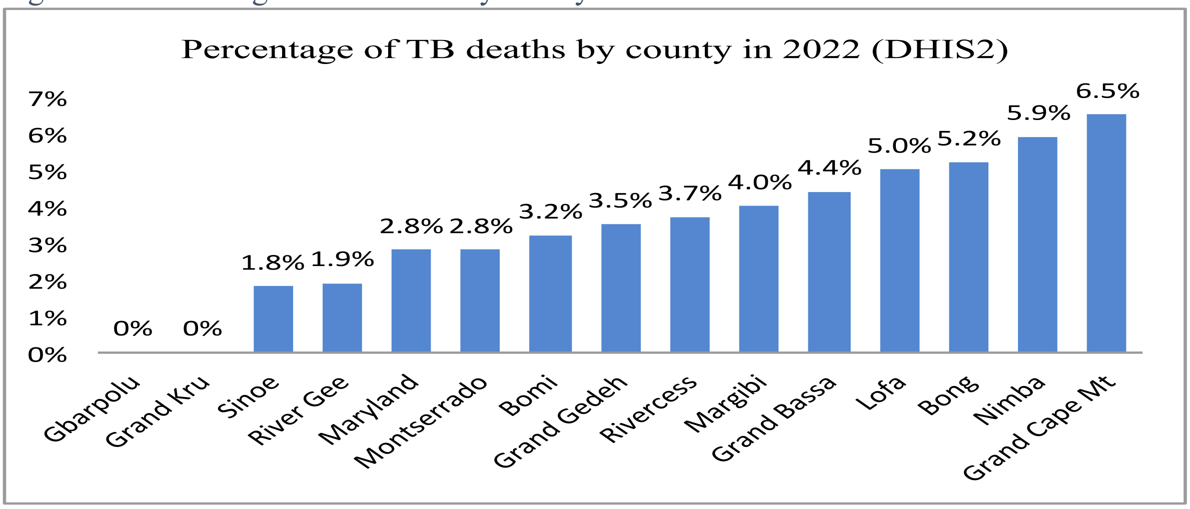


TB related deaths

Tuberculosis can be a life-threatening disease if left untreated or if the treatment is not completed. TB can cause severe damage to the lungs and other organs in the body, leading to complications and, in some cases death.

However, TB is treatable with appropriate antibiotics, and the majority of cases can be cured if the treatment is started early and followed through to completion. Public health efforts, including TB control programs and increased awareness, have been effective in reducing TB-related deaths in many parts of the world. It's important to seek medical attention promptly for diagnosis and appropriate treatment to improve the chances of a successful recovery and reduce the risk of death. In 2022, Gbarpolu and Grand Kru Counties did not report any TB related death. The highest deaths were reported in Grand Cape Mt (6.5%), Nimba (5.9%) and Bong (5.2%). Figure 36 presents the percentage of TB deaths by county in 2022.

Figure 36: Percentage of TB deaths by county in 2022



3.5.3 HIV

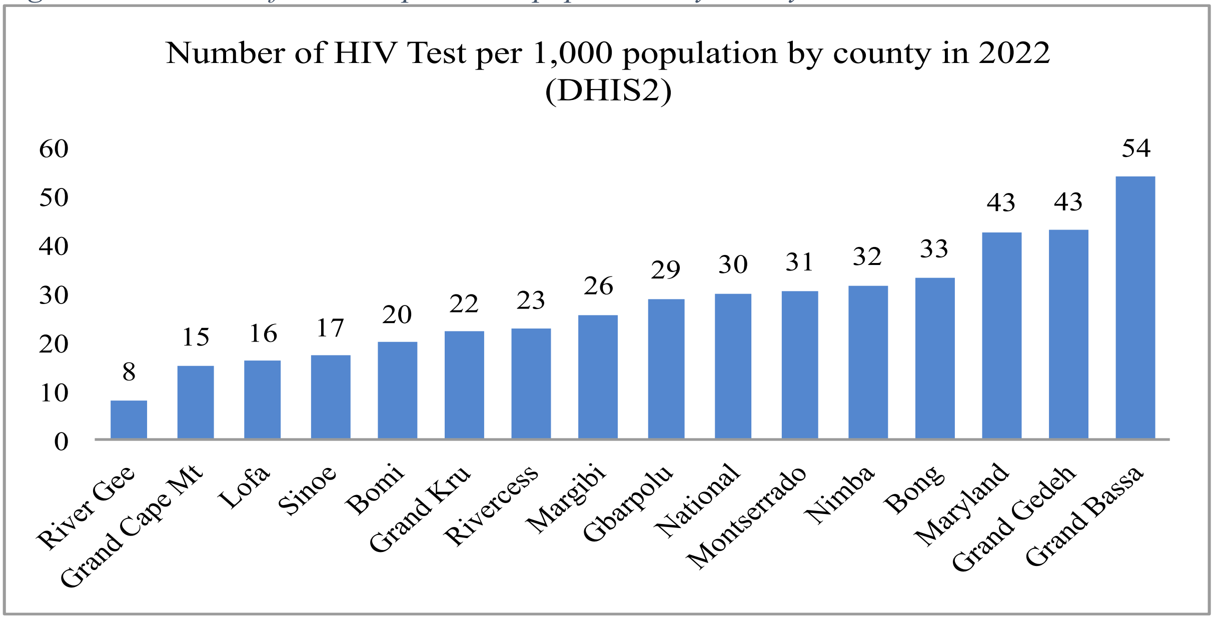
HIV or Human Immunodeficiency Virus, is a virus that attacks the immune system, specifically the CD4 cells (T cells), which help the immune system fight off infections. If left untreated, HIV can lead to AIDS (Acquired Immunodeficiency Syndrome), a condition in which the immune system is severely damaged. HIV is primarily transmitted through unprotected sexual contact, sharing needles for drug use, or from mother to child during childbirth or breastfeeding. It's important to get tested for HIV and, if necessary, seek medical treatment to manage the virus and prevent its progression.

HIV Testing

HIV testing is a crucial step in knowing your HIV status and taking appropriate measures to protect your health and prevent the spread of the HIV virus. It's important to get tested regularly, especially if you engage in high-risk behaviors or have had potential exposure to HIV. Testing is confidential, and many hospitals and health centers, and healthcare providers offer HIV testing services. If you test positive, it's crucial to seek medical care and follow the advice of healthcare professionals to manage the virus effectively.

Grand Bassa County (54 tests per 1,000 population) conducted the highest number of HIV tests per 1,000 population followed by Grand Gedeh (43 tests) and Maryland (43 tests) Counties. The lowest number of tests was conducted by River Gee (8 tests) and Grand Cape Mt County (15 tests). Figure 37 presents the number of HIV tests per 1,000 population by county in 2022.

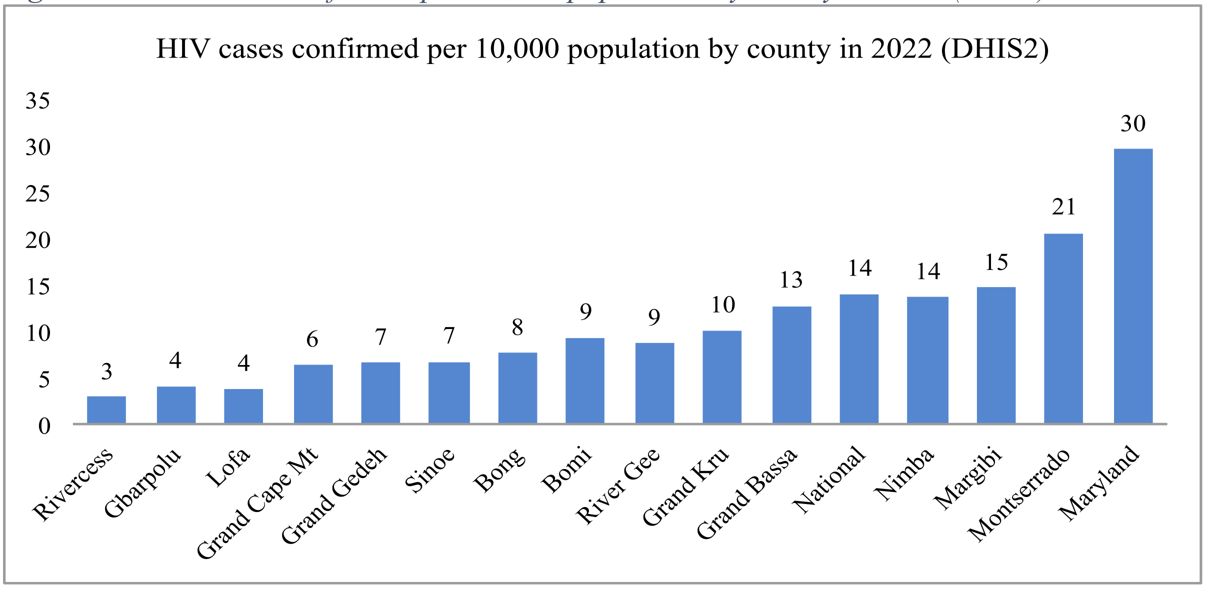
Figure 37: Number of HIV test per 1,000 population by county in 2022



HIV cases confirmed

The number of confirmed HIV cases can vary by location and change over time. The number of cases per 1,000 population was confirmed in Maryland County (30 cases), followed by Montserrado (21 cases) and Nimba (14 cases). The lowest number of confirmed HIV cases were reported from Rivercess (3 cases), Gbarpolu (4 cases), and Lofa (4 cases). Figure 38 displays the number of confirmed HIV cases in 2022 by county.

Figure 38: HIV cases confirmed per 10,000 population by county in 2022 (HMIS)



3.5.4 Childhood mortality

Childhood mortality, including under-five mortality and infant mortality, has been a significant concern in Liberia, particularly due to various health challenges and limited access to quality healthcare services.

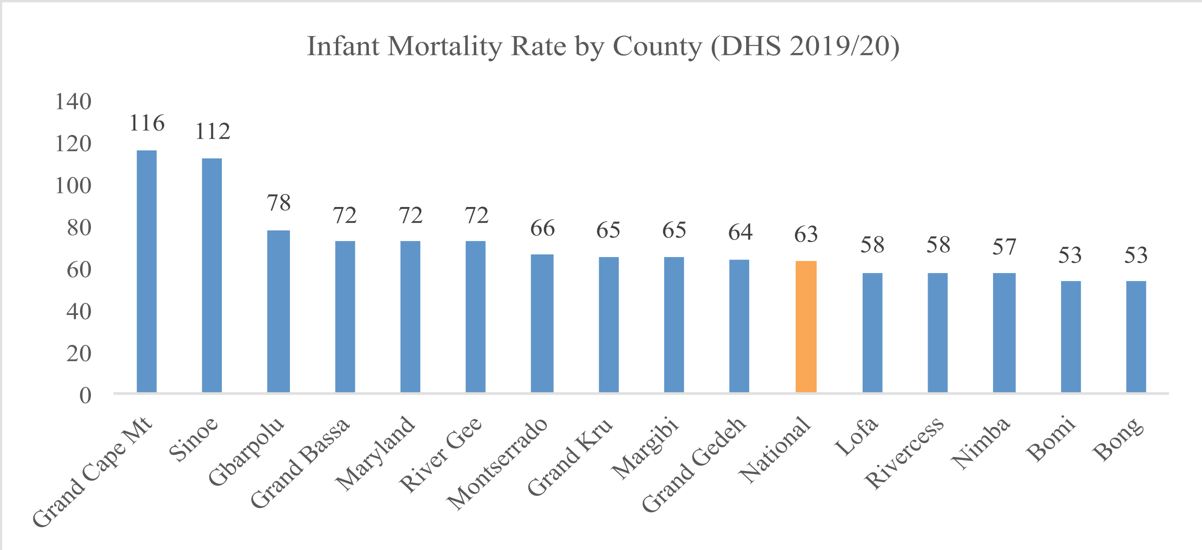
Liberia's under-five mortality rate, which measures the probability of a child dying before reaching the age of five, is relatively high. Various factors contributed to this, including limited access to healthcare services, malnutrition, and infectious diseases like malaria and diarrhea diseases.

Infant mortality, which measures the probability of a child dying during the first year of life, is also a concern. Lack of access to skilled birth attendants, maternal malnutrition, and preventable diseases played a role in infant mortality.

Common causes of childhood mortality included malaria, acute respiratory infections, diarrhea disease, malnutrition, and vaccine-preventable diseases.

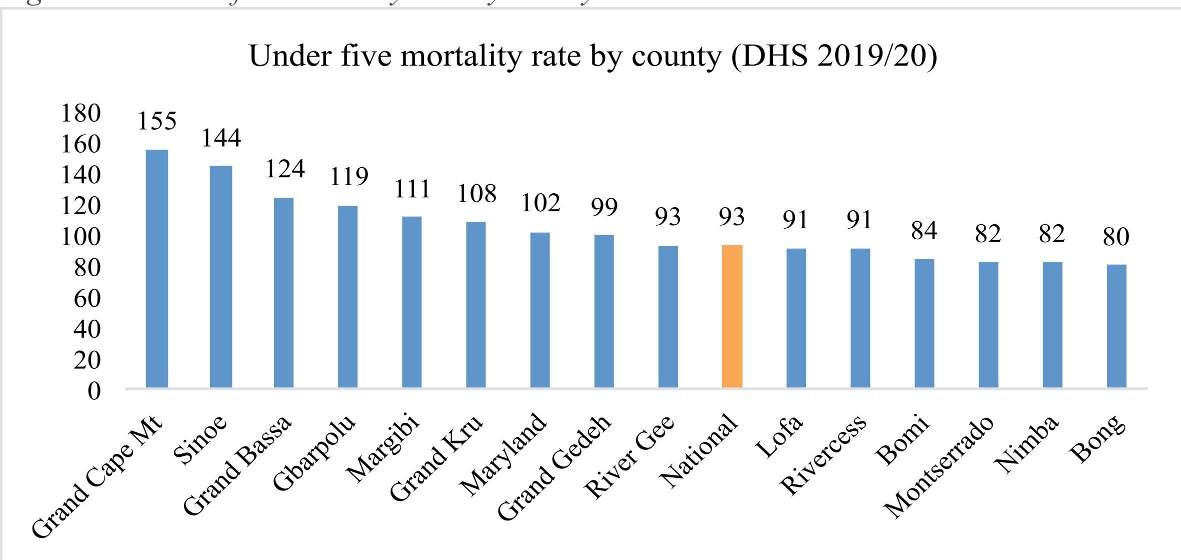
Liberia is among countries with high infant mortality rate. The current rate is 63 deaths per 1,000 live births. The highest infant mortality rates are found in Grand Cape Mount County (116 deaths per 1,000 live births), and Sinoe County (112 deaths per 1,000 live births). Also, the lowest infant deaths were recorded in Bong County (53 deaths per 1,000 live births), and Bomi County (53 deaths per 1,000 live births). Figure 39 shows infant mortality rate by county in Liberia.

Figure 39: Infant Mortality Rate by County



Liberia has one of the highest under five mortality rates in the world. The current rate is 93 deaths per 1,000 live births. There are disparities in under five mortality rates across the 15 counties of Liberia. Eight out of 15 counties have a higher under five mortality rate compared to the national rate. Grand Cape Mount has the highest rate of 155 deaths per 1,000 live births followed by Sinoe County (144 deaths per 1,000 live births), and Grand Bassa County (124 deaths per 1,000 live births). Counties with low under five deaths compared to the national level are, Bong (80 deaths per 1,000 live births), followed by Nimba (82 deaths), and Montserrado (82 deaths). Figure 40 presents under five mortality rate by county[21].

Figure 40: Under five mortality rate by county

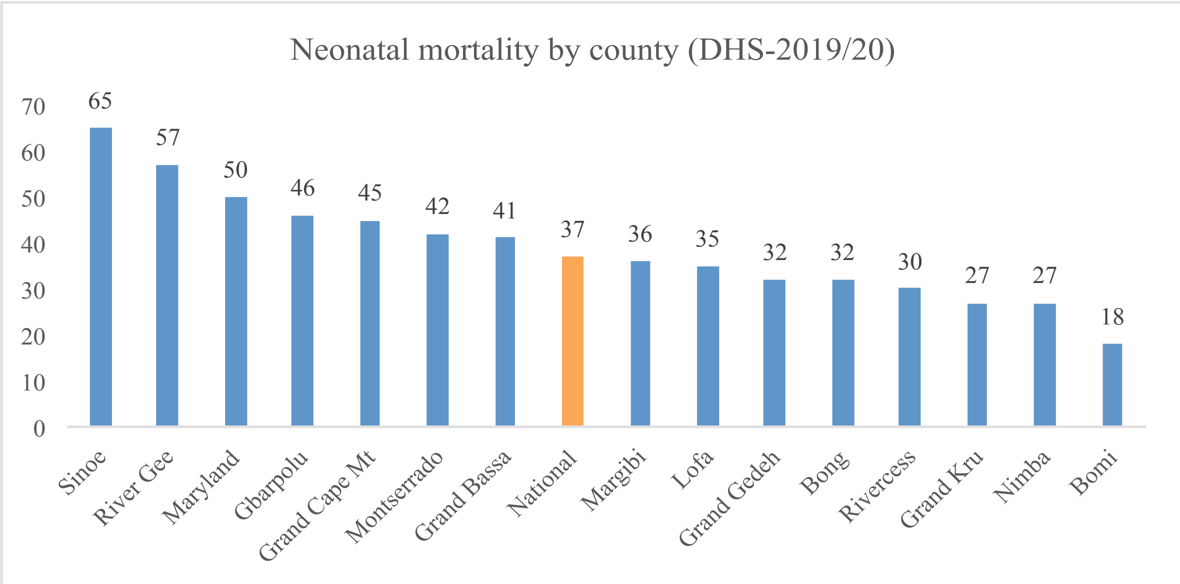


Neonatal mortality in Liberia is a significant public health concern. Liberia has a high neonatal mortality rate, with approximately 37 neonatal deaths per 1,000 live births[22]. Factors contributing to high rate includes; limited access to quality healthcare services, insufficient

[21] Ibid
[22] Liberia Demographic and Health Survey, 2019/20

maternal and child healthcare infrastructure, and challenges related to poverty and education. Neonatal mortality rate varies from county to county. Sinoe County has the highest neonatal mortality of 65 deaths per 1,000 live births, followed by River Gee County (57), and Maryland County (50 deaths per 1,000 live births). Counties with the lowest neonatal mortality rates are, Bomi (18), Nimba (27) and Grand Kru (27). Figure 41, depicts neonatal mortality rates by county in 2022[23].

Figure 41: Neonatal mortality rate per 1,000 live births by county in 2022



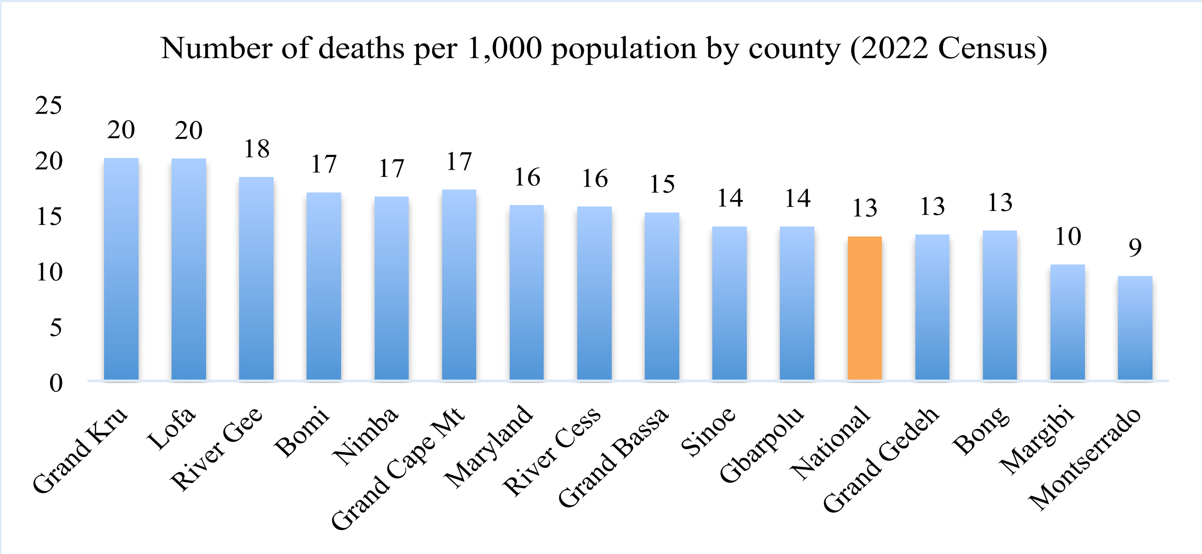
3.5.6 Adult Mortality

Adult mortality refers to the number of deaths among individuals in the adult population, typically defined as those aged 15 to 64years, within a specific time frame. It is an important demographic and health indicator that provides insights into the overall health and well-being of the population. Factors such as disease prevalence, healthcare access, and socioeconomic conditions can influence adult mortality rates in a given location.

The number of adult deaths varies across Liberia with Grand Kru County (20 deaths per 1,000 population), and Lofa County (20 deaths) having the highest number of deaths per 1,000 population compared to Montserrado (9) and Margibi (10) Counties with the lowest death rates. Figure 42 presents the number of deaths per 1,000 population by county in 2022[24].

[23] Ibid
[24] National Population and Housing Census, 2022

Figure 42: Number of deaths per 1,000 population by county in 2022



Maternal mortality in Liberia has been a significant public health concern. Liberia had one of the highest maternal mortality rates in the world. Several factors contributed to the issue:

a) Many women in Liberia lacked access to essential healthcare services, especially in remote and rural areas. The limited access to prenatal care, skilled birth attendants, and emergency obstetric care.

b) Inadequate healthcare infrastructure and facilities, including a shortage of well-equipped hospitals and clinics, contributed to the challenges faced by pregnant women.

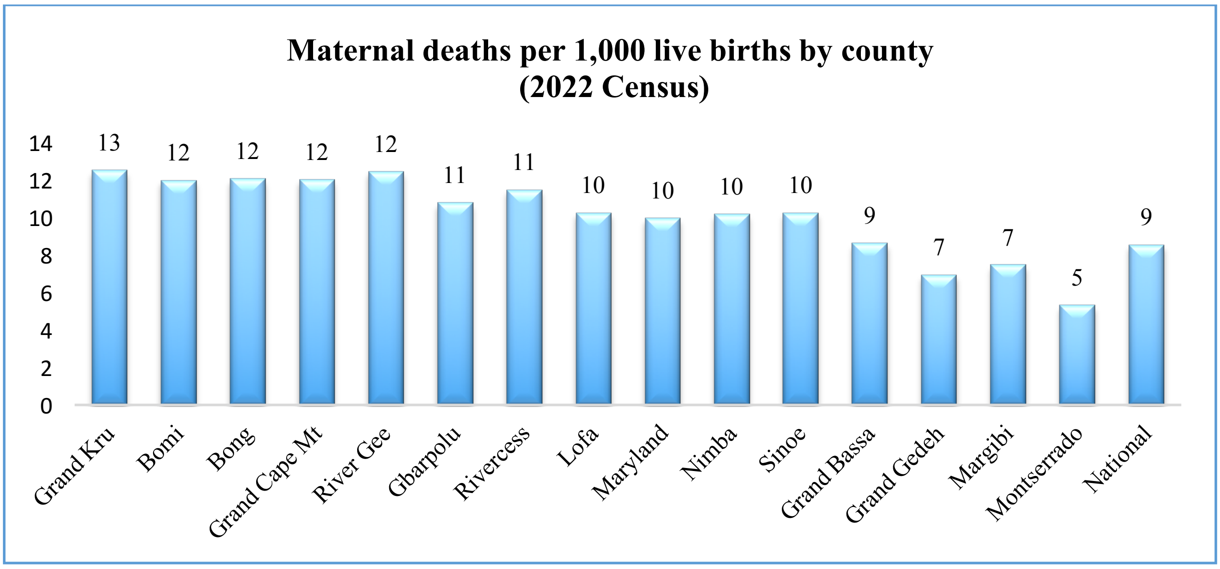
c) Poverty, limited education, and cultural factors often played a role in preventing women from seeking timely medical care during pregnancy and childbirth.

d) High fertility rate in Liberia also increased the risk of maternal mortality as women had a higher likelihood of experiencing multiple pregnancies.

Maternal death is high in one-third of the counties (Grand Kru, Bomi, Bong, Grand Cape Mt and River Gee) and lower in three out of 15 counties (Montserrado, Margibi and Grand Gedeh). Figure 43 shows maternal deaths per 1,000 live births by county[25].

[25] Ibid

Figure 43: Maternal deaths per 1,000 live births by county (2022 Census)



Efforts are being made to address maternal health through initiatives aimed at improving healthcare infrastructure, increasing access to family planning, and raising awareness about maternal health. A comprehensive Reproductive, Maternal, Newborn, Child and Adolescent (RMNCAH) Strategy is been implemented to improve antenatal care, obstetric emergencies, introduction of task shifting, provision of basic equipment, and development of service providers' skills and knowledge.

3.6 Human Resources for Health (HRH)

Human resources for health (HRH) in Liberia refers to the healthcare professionals such as medical doctors (MD), nurses, physician assistants, midwives, Laboratory technicians, and other medical staff, who provide essential health services in the country. Liberia, like many developing nations, faces challenges in HRH, including shortage of trained healthcare workers, mal-distribution of staff between urban and rural areas, and inadequate training facilities.

Efforts to strengthen HRH in Liberia typically involve initiatives like:

a) Expanding medical training institutions and programs to produce more healthcare professionals.
b) Incentivizing healthcare workers to serve in underserved areas through programs like rural posting allowances.
c) Developing and implementing policies and regulations to govern the practice and licensing of healthcare professionals.

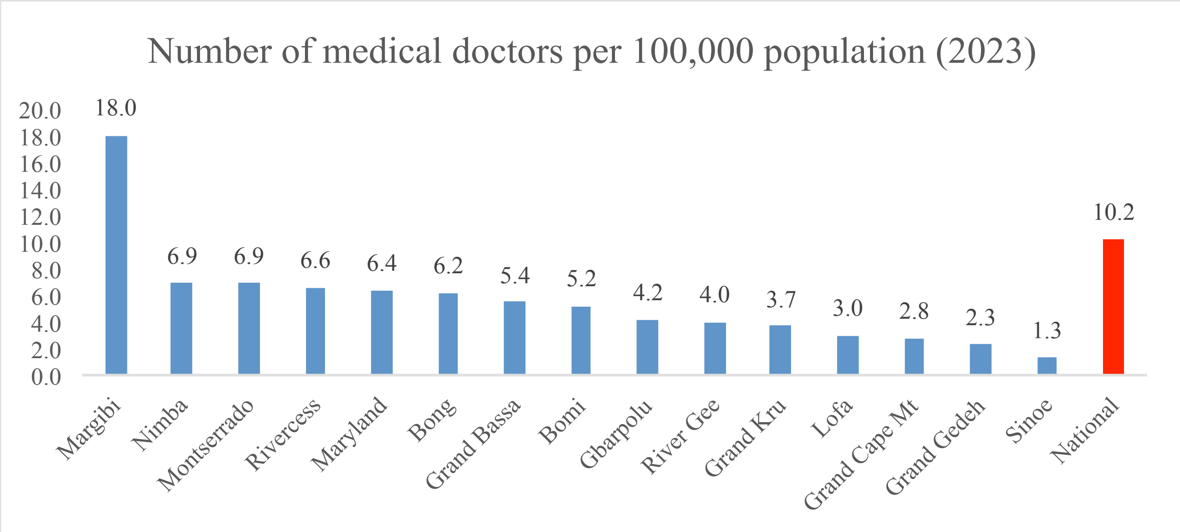
Liberia has made progress in recent years in strengthening its HRH, but it's an ongoing challenge that requires continued investment and focus to meet the healthcare needs of its population.

3.6.1 Medical Doctor Density

Medical doctor density refers to the number of medical doctors per capita in a specific area, often measured as the number of doctors per 1,000 or 10,000 people. It is an important indicator of a county's healthcare resources and its ability to provide medical services to its population. The density can vary significantly from one county to another, impacting access to healthcare services.

The patients to doctor ratio is low due to insufficient doctors in Liberia to provide services to a population of 5.2 million. The doctor to population ratio shows a very low ratio in Sinoe (1.3 doctors to 100,000 population), Grand Gedeh (2.3) and Grand Cape Mount Counties. Margibi has the highest doctor to population ratio. Figure 44 presents the number of medical doctors per 100,000 population by county[26].

Figure 44: Number of medical doctors per 100,000 population



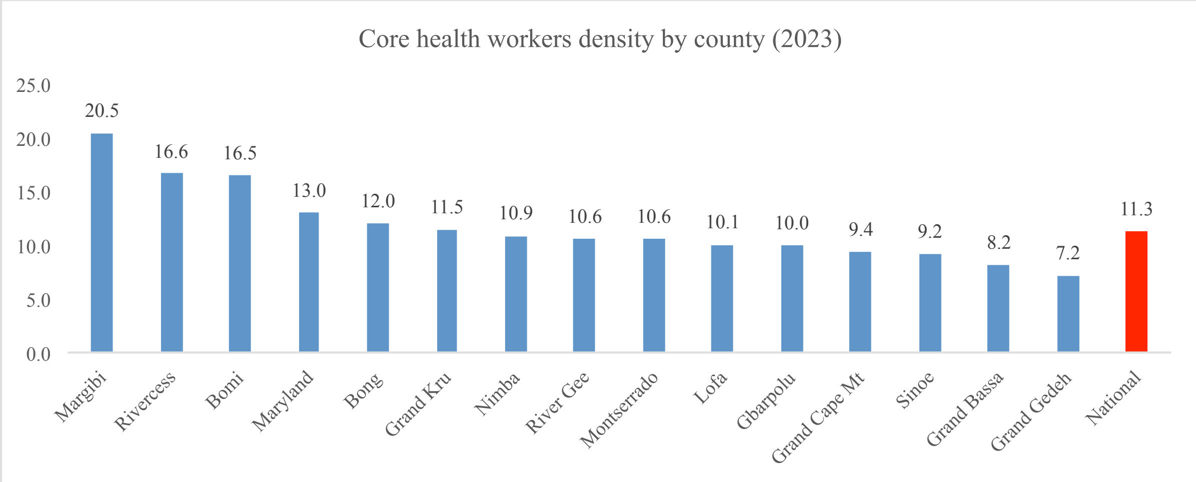
3.6.2 Core Health Workers Density

Core health worker density refers to the number of essential healthcare professionals, such as doctors, nurses, and midwives, per specific population or geographical area. It's a crucial metric for assessing a county's capacity to provide healthcare services. To calculate core health worker density, you would typically divide the number of healthcare workers by the population of the area in question. This metric helps evaluate healthcare access and can inform healthcare policy and resource allocation decisions.

The population to core workforce varies from one county to another. The disparity across counties is visible and huge. Margibi County has the highest core health worker density of 20.5 per 10,000 population while Grand Gedeh has the lowest density ratio of 7.2 per 10,000 population. Figure 45 shows the core health workers density by county.

[26] MOH Health Workforce Mapping Exercise, 2023

Figure 45: Core health workers density by county

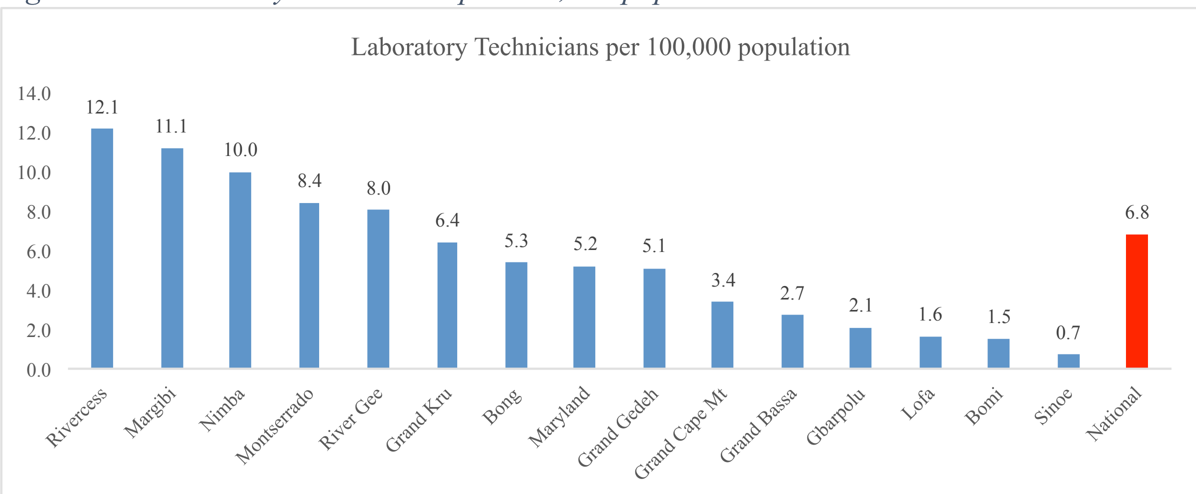


3.6.3 Laboratory Technicians Density

Laboratory technician density is a metric that measures the number of laboratory technicians or technologists in a specific area or healthcare system relative to the population it serves. Lab technicians play a critical role in diagnostic testing, research, and medical investigations. To calculate lab technician density, you would divide the number of lab technicians in an area of facility by the population or patient base they serve. This metric can be important for assessing a county's ability to conduct essential laboratory tests and support medical services. It's particularly relevant in times of public health crises, such as the COVID-19 pandemic, where diagnostic testing is crucial.

There is an acute shortage of Laboratory technicians in Liberia. The health workforce mapping data of 2023 displays significant variations from one county to another. The Lab technician density ratio shows less than 1 Lab technician per 100,000 population in Sinoe County. On the other hand, there are 12.1 Lab technicians and 11.1 in Rivercess and Margibi Counties respectively. Figure 46 depicts Laboratory Technicians per 100,000 population by county[27].

Figure 46: Laboratory Technicians per 100,000 population



[27] MOH Health Workforce Mapping exercise, 2023

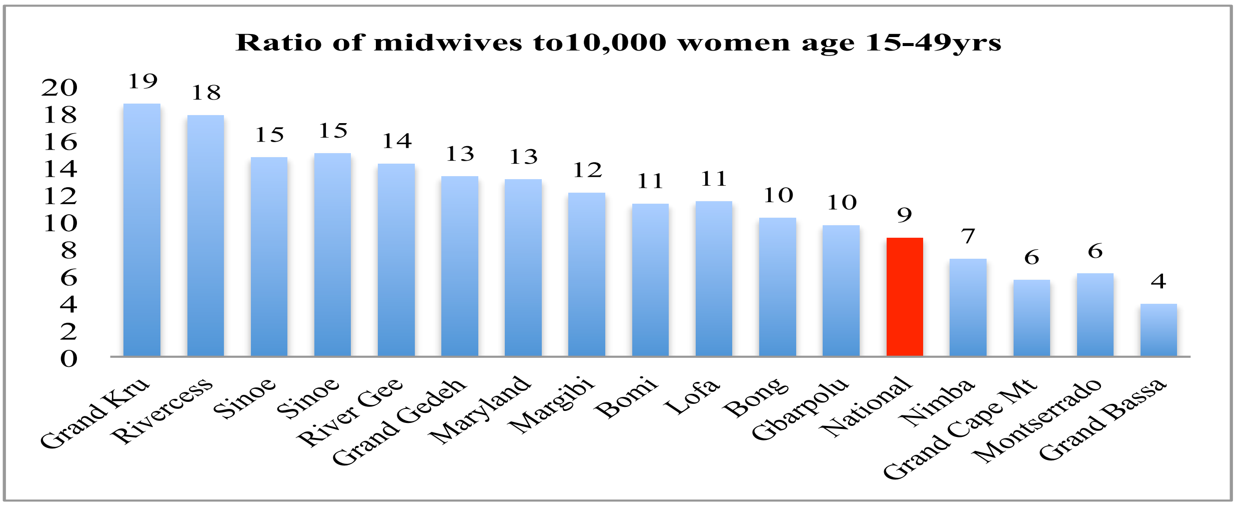
3.6.4 Midwives Density

Midwives are skilled healthcare professionals specialized in assisting women during pregnancy, childbirth, and the postpartum period. They offer support and monitor the health of both the mother and the baby.

Midwives play a crucial role in women's healthcare by providing personalized and holistic care throughout the childbirth process. Their expertise in supporting natural birthing process, promoting maternal health, and offering individualized care, contributes significantly to positive birth experiences and overall maternal well-being.

In Liberia, the number of midwives per county varies, with the Southeastern counties having higher midwives to women of reproductive age (15-49) ratio than others regions. Grand Kru County has the highest ratio of 19 midwives to 10,000 women of reproductive age. The lowest ratio is found in Grand Bassa (4 midwives to 10,000 women age 15-49yrs)[28]. Figure 47 below shows the ratio of midwives to women of reproductive age across Liberia.

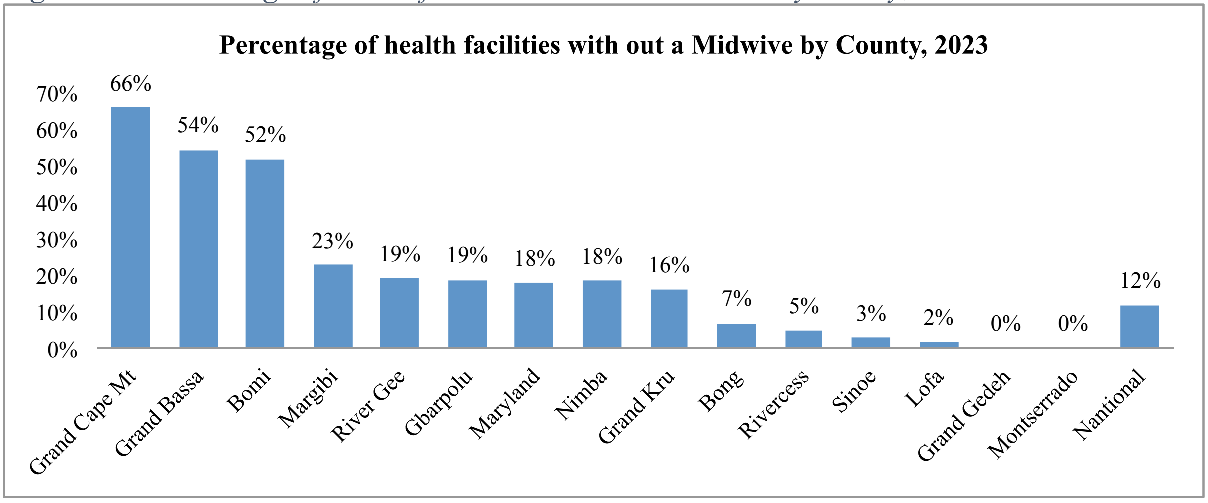
Figure 47: Ratio of midwives to 10,000 women of reproductive age, 2023



The health workforce assessment found that 12% of the 935 health facilities in the country do not have midwives to provide care for women of reproductive age. Two-thirds of health facilities in Grand Cape Mount County and over half of health facilities in Grand Bassa (54%) and Bomi (52%) do not have assigned midwives. Figure 48 presents the percentage of health facilities with no midwives by county[29].

[28] MOH Health workforce mapping exercise, 2023
[29] MOH Health Sector Resource Mapping Exercise, 2023

Figure 48: Percentage of health facilities with out midwives by county, 2023



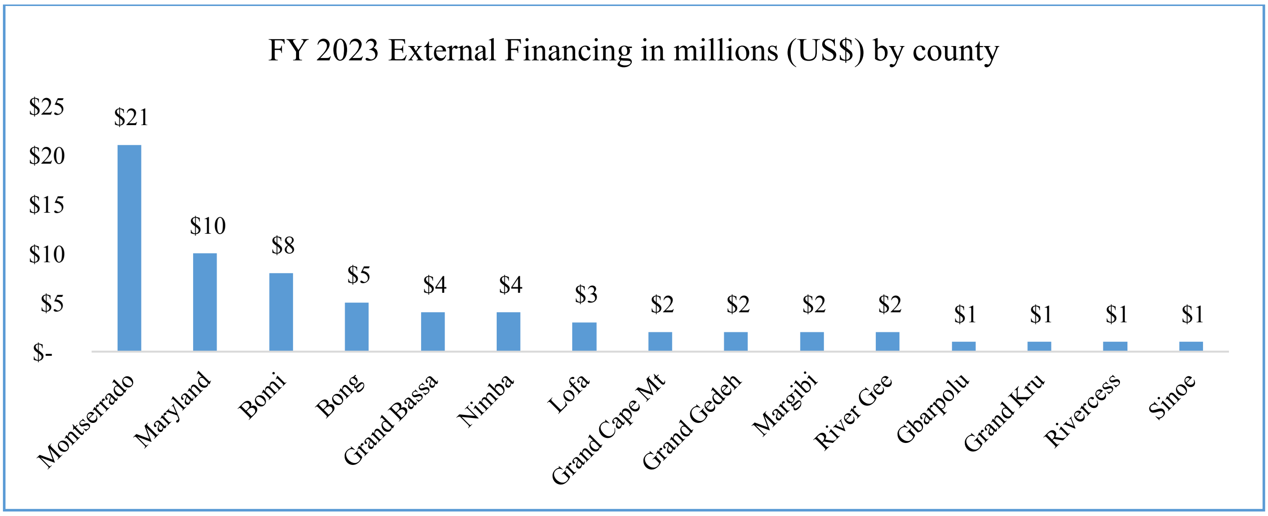
3.7 Health expenditures

Liberia has relatively low healthcare expenditure that is faced with significant challenges due to external shocks and low revenue generation capacity. The government and international organizations are working to improve healthcare services by instituting Performance Based Financing (PBF), Fixed Asset Reimbursement Arrangement (FARA) among others. However, financial allocation (domestic and external) varies greatly by counties and health programs.

3.7.1 External Financing

In fiscal year (FY) 2023, the 15 counties are to benefit US$68.3 million (47%) of the total external investment of $45.5 million in the Liberian health sector. As shown in Figure 34, Montserrado, Maryland, Bomi, and Bong Counties are expected to get the largest proportion of total foreign investments of US$21 million, US$10 million, US$8 million, and US$5 million, respectively[30]. Figure 49 presents fiscal year 2023 external financing by county in millions (US$).

Figure 49: Fiscal year 2023 external financing in millions (US$) by county



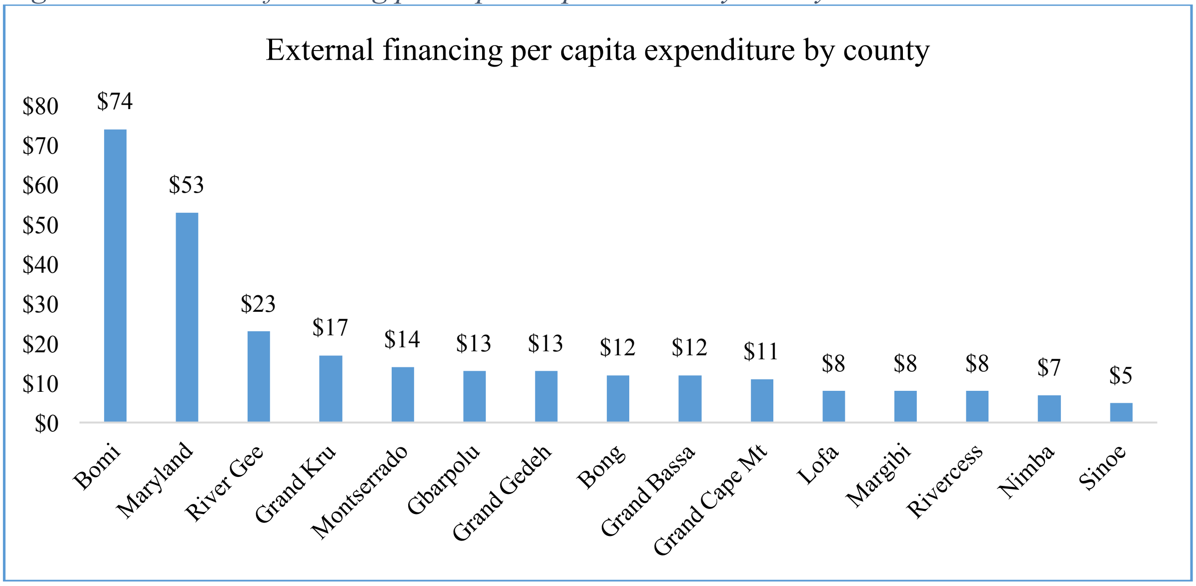
[30] Ibid

3.7.2 External health expenditure per capita

External health expenditure per capita refers to the amount of money spent on healthcare for each person in a given country that comes from sources outside the country. This can include foreign aid, donations, and international funding for health programs. It varies significantly from one county to another.

When total foreign investment in each county is divided by population, Bong County is predicted to benefit the most from US$74 in investment per person, followed by Maryland and River Gee Counties, which are expected to benefit from US$53 and US$23, respectively. Figure 50 depicts the external financing per capita expenditures by county.

Figure 50: External financing per capita expenditures by county

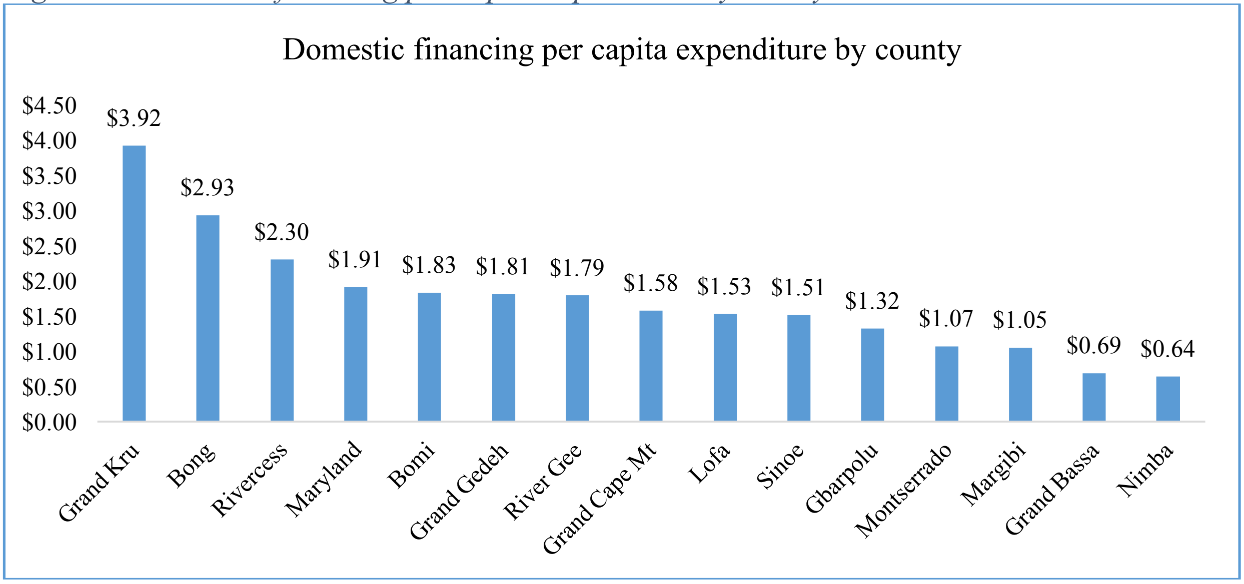


3.7.3 Domestic health expenditure per capita

Domestic health expenditure per capita refers to the average amount of money spent on healthcare for each person within a specific country. This expenditure includes both public and private spending on healthcare services such as hospitals, doctors' visits, medications, and more. The domestic health expenditure per capita varies greatly from one county to another and are influenced by factors like the county's healthcare system, the overall health of the population, and the level of economic development.

There is disparity in the allocation of domestic resources across Liberia. Grand Kru County has the highest amount (US$ 3.92 per person) of domestic financing per capita, followed by Bong (US$ 2.93 per persons) and Rivercess Counties (US$ 2.30). Also, Nimba County has the lowest (US$ 0.64) per capita health expenditure, followed by Grand Bassa County (US$ 0.69). Figure 51 displays the amount of domestic financing per capita.

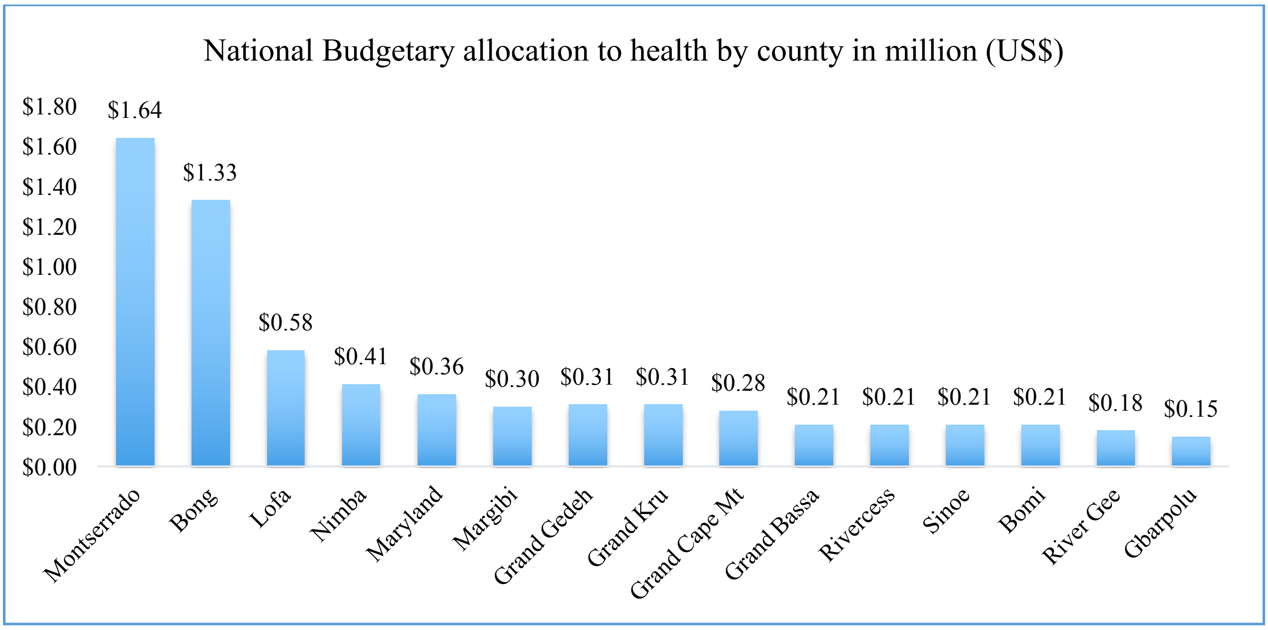
Figure 51: Domestic financing per capita expenditure by county



3.7.4 Budgetary allocation

Budgetary allocation refers to the process of assigning financial resources or funds to specific government ministry and agency, programs, departments, units, or projects within the government's budget. It involves deciding how public funds are distributed among various sectors. Government's budgetary allocation vary from county to county within the health sector, with Montserrado County having the highest share (US$ 1.64 million), followed by Bong (US$ 1.33 million). Counties with the lowest share of the budget are, Gbarpolu (US$0.15 million), followed by River Gee (US$ 0.18 million). Figure 52 depicts the national budgetary allocation to health by county in million (US$).

Figure 52: National budgetary allocation to healthcare by county in million (US$)



4.0 Conclusion and Recommendations

4.1 Conclusion

Liberia faces a complex interplay of social determinants that significantly impact the health and well-being of the population. These social determinants encompass various economic, political, cultural, and environmental factors. For instance, a significant portion of Liberia's population lives below the poverty line, leading to limited access to health care, nutritious food, and safe housing.

The impact of high unemployment rates on financial stability and access to necessary health services is deeply concerning. Woefully, Liberia is experiencing this issue on a large scale, with a significant portion of the population aged 15 and above currently unemployed.

Unfortunately, many Liberians do not have access to formal education, with a significant portion of females and males aged six years and older having no formal education or only some elementary education.

Addressing social determinants of health in Liberia requires multi-sectoral efforts, including investments in healthcare infrastructure, education, poverty reduction, gender equality, and environmental sustainability. However, Liberia has made some progress in gender equality by electing the first female president in Africa and has appointed women to senior government positions (e.g.: Ministers, CEO, Director General, etc) and elected women to the national legislature.

Efforts to address health equality and inequality involve policies and initiatives that aim to reduce disparities, improve healthcare access, and promote social determinants of health. These strategies can include providing affordable healthcare, reducing income, and inequality, improving education, and addressing systemic factors that contribute to health disparities. Addressing social determinants of health in Liberia requires multi-sectoral efforts, including investments in healthcare infrastructure, education, poverty reduction, gender equality, and environmental sustainability.

Liberia has made progress in recent years in strengthening its HRH, but it's an ongoing challenge that requires continued investment and focus to meet the healthcare needs of its population. The patient-to-doctor ratio is low in Liberia due to insufficient doctors in Liberia to provide services to a population of 5.25 million.

There is an acute shortage of Laboratory technicians in Liberia. The Lab technician density ratio shows less than 1 Lab technician per 100,000 population in two counties, while on the other hand, two counties have over 12 Lab technicians per 100,000 population.

Investment in healthcare is relatively high but most of the investment is not aligned to the health sector priorities. There is a disparity in the allocation of domestic resources across Liberia with few counties having higher per capita health expenditure than others.

Since the 2020/21 fiscal year (FY), the national budget continues to increase. The budget increased from US$ 570.1 million in FY 2020/21 to US$ 782.9 million in 2023. On the contrary, external funding to the health sector continues to decline from US$ 339 million in FY 2017/18 to US$ 147.60 million in 2023.

The majority of (88%) urban dwellers have access to health care compared to 49% of rural dwellers. However, health services utilization is very low in Liberia. Over the past 5years (2018-2022), utilization has been below one visit per capita which is far below the WHO recommended five visits per capita per year.

Liberia's healthcare infrastructure is limited and faces significant challenges. Many health facilities, particularly in rural areas, are in need of improvement in terms of equipment, staffing, and access to essential services. Additionally, there is an insufficiency of hospital beds, as a result, patients are either prematurely discharged or refused admission.

The proportion of children 12-23 months that have received all basic vaccines varies from county to county with Sinoe County having little over a quarter (27%) of their children receiving all basic vaccines.

Most of the deliveries occur in healthcare facilities. However, there are disparities across counties. The proportion of health facility delivery is highest in Lofa (96%), Nimba (92%), and River Gee (91%). Nearly 90% of health facilities provide BEmOC services.

Mortality is increasing among all sub-groups of the population. For instance, over the past three years, on average five maternal deaths and 15 neonatal deaths have been reported weekly. To address this appalling and undesirable situation, the government and its development partners need to invest in the quality of maternal and newborn health services, ensure the availability of essential drugs and life-saving commodities, improve referral services, and expand CEmOC facilities and their quality.

4.2 Recommendations

Interventions to address health equality and inequality involve policies and initiatives that aim to reduce disparities, improve healthcare access, and promote social determinants of health. These strategies must include providing affordable healthcare, improving income, and education, and addressing systemic factors that contribute to health disparities. Most of the contributing factors to address health equality and inequity occur beyond the health sector. It is important to strengthen multi-sectoral and multi-disciplinary advocacy and collaboration.

Achieving health equity is a complex and ongoing challenge, but it is essential for creating a just and healthy society.

Below are recommended actions to improve health equity and equality:

1). Government of Liberia

1. To combat stunting, a multi-pronged approach is needed. This includes improving maternal nutrition, promoting breastfeeding, ensuring access to nutritious foods, enhancing sanitation and hygiene, and providing proper healthcare.
2. To increase physical access to healthcare and reduce disparities across counties, the government of Liberia needs to invest in the construction of primary health clinics in underserved communities especially in Bong, Gbarpolu, Grand Bassa, Nimba, and Grand Gedeh among others.

3. To improve the low utilization of health services and poor quality of health care, the government and development partners need to invest and ensure the availability of essential drugs and medical supplies at public health facilities, deploy basic equipment and diagnostic equipment, and introduce a social health insurance scheme.
4. To address the critical shortage of health workforce such as laboratory technicians, certified midwives, medical doctors, specialists, and other cadres, the government and development partners must improve and expand health training institutions, as well as ensure equitable distribution of health workers and training institutions.
5. Resource allocation to counties must be done equitably and consider population, disease burden, geography, and the status of health infrastructure,
6. Government must strive to ensure equity distribution of resources, and reduce off-budget expenditures.
7. A concerted effort and explicit actions are required to improve the socioeconomic determinants of health that have implications for the population's health and well-being. These determinants include; improvement in women's education, road connectivity, sources of drinking water, electricity, toilet facilities, sanitation, and households' income.

2). Development partners

1. Partners must allocate resources to counties based on the following considerations; population, disease burden, the status of health infrastructure, and ensure equity distribution.
2. Partners must reduce off-budget expenditures and align their resources to the national priorities and initiatives.
3. Partners should consider and adhere to the International Health Partner Plus (IHP+) framework.
4. MOH should conduct a research on the correlation between health investments and health outcomes.

3). Civil Society Organizations (CSOs) and Non-Government Organizations (NGOs)

1. Efforts to promote good governance and reduce corruption are essential for improving health outcomes and overall well-being in Liberia.
2. CSOs need to monitor the quality of health services provide feedback to the government and promote healthy lifestyles and well-being
3. NGOs need to align their resources with the national health policy and plan and improve coordination and collaboration with the Ministry of Health to reduce fragmentation and duplication.
4. CSOs and NGOs need to promote women's education and empowerment to reduce the impact of social determinants on their health and well-being.

4). Community

1. MOH must involve communities in the promotion of quality healthcare delivery
2. MOH and partners must engage the community to take charge of their own health and monitor health service delivery.
3. MOH should strengthen community health structures to provide oversight and own health service delivery at the community level.

4. MOH and partners should improve community demand generation, enhance service utilization and promote gender equity and women empowerment initiatives.

References

LISGIS: Household Income and Expenditure Survey, 2016

LISGIS & USAID: Liberia Demographic and Health Survey, 2019/2020

LISGIS: Liberia Population and Housing Census, 2022

Ministry of Health & World Health Organization: Harmonized Health Facility Assessment, 2021

Ministry of Health & World Health Organization: Health workforce mapping exercise, 2023

Ministry of Health: Health Management Information System (HMIS/DHIS-2) data, 2022

Ministry of Health: List of Accredited Paramedic Schools, 2023

Annex A: Health Services Utilization by County (2018-2022)

County	**2018**	**2019**	**2020**	**2021**	**2022**
Bomi	1.17	0.85	0.82	0.81	0.75
Bong	0.90	0.74	0.67	0.66	0.61
Gbarpolu	0.63	0.52	0.52	0.46	0.49
Grand Bassa	0.92	0.84	0.68	0.70	0.70
Grand Cape Mount	0.80	0.68	0.69	0.66	0.61
Grand Gedeh	1.07	0.84	0.80	0.76	0.76
Grand Kru	1.32	0.98	1.05	0.99	0.98
Lofa	1.03	0.81	0.76	0.68	0.65
Margibi	0.97	0.85	0.66	0.74	0.77
Maryland	1.13	0.87	0.77	0.86	0.80
Montserrado	1.21	1.02	0.83	0.94	0.87
Nimba	0.88	1.01	0.70	0.65	0.58
Rivercess	1.34	0.69	0.76	0.64	0.55
River Gee	1.34	0.99	1.06	0.99	0.77
Sinoe	1.43	1.01	0.86	0.82	0.59
National	**1.06**	**0.90**	**0.76**	**0.79**	**0.73**

Annex B: List of Accredited Nursing and Midwifery Schools

List of Nursing and Midwifery Schools Assessed and Met the Liberian Board for Nursing and Midwifery's Standards		
#	**Name of Training Institution**	**Location**
1	Tubman National Institute of Medical Arts (TNIMA)	Montserrado County
2	Winifred J. Harley College of Health Sciences	Nimba County
3	Cuttington University Nursing college	Bong County
4	Phebe School of Nursing and Midwifery	Bong County
5	Esther Bacon School of Nursing and Midwifery	Lofa County
6	Dianna Kay Isaacon Training Program	Grand Gedeh County
7	Mother Patern College of Health Sciences	Montserrado County
8	Christain O. Symthe College of Health Sciences	Montserrado County
9	Lois Hemgren School of Nursing	Montserrado County
10	William V. S. Tubman University College of Health Sciences	Maryland County
11	Grand Bassa Community College of Health Sciences	Grand Bassa County
12	Mabel McComb School of Health Sciences	Montserrado County
13	Ruth Ramstrand School of Nursing	Lofa County
14	United Methodist University (UMU) Advanced Midwifery Program	Montserrado County
15	Bomi Community college of Nursing	Bomi County
16	Lofa Community College of Nursing	Lofa County
17	Seventh Day Adventist University School of Nursing	Margibi County
18	Nimba County Community College of Nursing	Nimba County
19	Cuttington University Junior College	Margibi County
20	Greater Vision College of Nursing	Montserrado County
21	New Sight Eye Center (NSEC) Ophthalmic Nursing Training Program	Montserrado County
22	Censil College	Montserrado County

Annex C: Number of malnourish Children under five admitted, relapse and died by county in 2022

County	Admission (OTP & IPF)	Relapse (OTP & IPF)	Deaths	% of Relapse	% of Deaths
Bomi	1,263	6	4	0.5%	0.3%
Bong	2,976	17	11	0.6%	0.4%
Gbarpolu	1,090	5	8	0.5%	0.7%
Grand Bassa	1,380	19	5	1.4%	0.4%
Grand Cape Mt	1,216	11	2	0.9%	0.2%
Grand Gedeh	1,039	4	9	0.4%	0.9%
Grand Kru	636	9	5	1.4%	0.8%
Lofa	2,391	8	23	0.3%	1.0%
Margibi	1,553	9	4	0.6%	0.3%
Maryland	1,091	37	21	3.4%	1.9%
Montserrado	7,208	149	193	2.1%	2.7%
Nimba	2,230	10	28	0.4%	1.3%
River Gee	521	1	4	0.2%	0.8%
Rivercess	659	1	9	0.2%	1.4%
Sinoe	550	12	5	2.2%	0.9%
Total	***25,803***	***298***	***331***	***1.2%***	***1.3%***

Annex D: List of Health Facility by County, Type and Ownership

#	*County*	*Health Facility*	*Public*	*Private*	*Hospital*	*Health Center*	*Clinic*
1	Bomi	29	24	5	1	0	28
2	Bong	58	42	16	3	2	53
3	Gbarpolu	16	15	1	1	0	15
4	Grand Bassa	35	26	9	3	1	31
5	Grand Cape Mt	35	32	3	1	3	31
6	Grand Gedeh	24	22	2	1	2	21
7	Grand Kru	25	23	2	1	4	20
8	Lofa	61	54	7	4	3	54
9	Margibi	62	25	37	3	10	49
10	Maryland	28	25	3	1	2	25
11	Montserrado	396	57	339	9	25	362
12	Nimba	87	53	34	5	5	77
13	Rivercess	21	19	2	2	1	18
14	River Gee	21	20	1	1	2	18
15	Sinoe	37	35	2	1	0	36
	Total	***935***	***472***	***463***	***37***	***60***	***838***